DEMOCRATIC REPUBLIC OF CONGO
MINISTRY OF WATER RESOURCES AND ELECTRICITY WATER PROJECT EXECUTION UNIT

ACCESS TO WATER AND SANITATION SERVICES IN THE DRC (PASEA)
IDA Credit N°73390-ZR

Individual consultant in charge of developing Terms of Reference for the acquisition of one or more Management Information System(s) (MIS) on Water, Hygiene and Sanitation (WASH) aspects for the four targeted provinces of the PASEA Project: Kwilu, Kasai, Kasai Central and Kasai Oriental

DIAGNOSTIC REPORT OF THE INFORMATION MANAGEMENT SYSTEM AND STUDY OF GOOD PRACTICES

By: MUHINDO KYANA Malka
Date: April 7, 2025
Contact : mkyana3@gmail.com, +243994914940

© April 2025

CONTENTS

INTRODUCTION....4
1. General context of the PASEA project....4
2. Objectives....5
3. Methodological approach....6
CHAPTER 1: DIAGNOSIS OF EXISTING SYSTEMS IN THE PROVINCES TARGETED....7
1. Mapping of key stakeholders and status of existing IMS in structures....7
2. Inventory of existing data....36
3. Preliminary roadmap....38
CHAPTER 2: REVIEW OF NATIONAL AND NATIONAL GOOD PRACTICES INTERNATIONAL....40
1. National Good Practice: mWater in DRC....40
1. Integrate the sustainability model from the design phase....40
2. Include the on and offline working mode when collecting data....41
3. Co-create the collection tools with stakeholders....41
2. National Information System for Water Supply and Sanitation (NISWSS)....41
3. RUWASA Service Delivery Management System....43
4. Recommendations of the Ministry in charge of digital technology in the design of SGIs in Ground floor....45
CHAPTER 3: PROPOSED MODEL OF THE INFORMATION IN THE WATER AND SANITATION SECTOR IN THE DRC....47
1. Design Assumption....47
2. High-level overview of the proposed unified IMS-EHA....49
3. Roadmap....50
I. APPENDICES....51

ACRONYMS AND ABBREVIATIONS

Ground floor	Democratic Republic of Congo
PASEA	Access to Water and Sanitation Services Program in the DRC
ETD	Decentralized territorial entities
CEP-O	Water Project Implementation Unit
CNAEHA	National Water, Hygiene and Sanitation Action Committee
SAR	Sanitation Directorate
DESP	Directorate of Care Facilities and Partnerships
DHSP	Hygiene and Public Health Department
DEVC	Life Education Department
DINAC	National Construction Directorate
DRE	Water Resources Branch
ONHR	National Office of Rural Hydraulics
SG MRHE	General Secretariat of the Ministry of Water Resources and Electricity
CSPP	Project and programme monitoring unit

REGIDESO	Water distribution authority
RPSPE	Provincial Water Utilities Board
DPS	Provincial Health Division
CPEDD	Provincial Environment and Sustainable Development Coordination
BHSP	Office of Public Health and Sanitation
CADEVC	Coordination of the branch of the Directorate of Education for Everyday Life
EPST	Primary, secondary and tertiary teacher
DPRHE	Provincial Water Resources and Energy Division
WASH	Water-hygiene-sanitation
SGI	Information Management System
ARSPE	Public Water Services Regulatory Authority
SINIEau	National Integrated Water Information System
DHIS2	District Heath Information System 2
ALF	End of open defecation
SWOT	Strengths, weaknesses, opportunities and threats
NFT	Non-revenue water
NWASCO	National Water and Sanitation Council (Zambia)
NISWSS	National Information System for Water Supply and Sanitation (Zambia)
RUWASA SDMS	RUWASA Service Delivery Management System
SDGs	Sustainable Development Goals
API	Application programming interface
PNN	National Digital Plan

Table 1: List of acronyms

TABLES AND FIGURES

Table 1: List of acronyms and acronyms 2
Table 2: Organisations covered by the ..... *diagnosis* 7
Table 3:CNAEHA critical data ..... Art. 10
Table 4: SWOT Analysis of IMS at NAFHAA ..... Art. 10
Table 5:Critical data SAR ..... Art. 14
Table 6: SWOT analysis of the SAR ..... Art. 14

Table 7: Existing IMS within the provincial directorates of ONHR ..... Art. 16
Table 8: Critical data from ONHR ..... Art. 17
Table 9: SWOT analysis of ONHRS ..... Art. 17
Table 10:Critical data from the DHSP 20
Table 11:SWOT analysis of the DHSP ..... Art. 21
Table 12:Critical data from the SG-MRHE 24
Table 13:SWOT Analysis of SG-MRHE ..... Art. 24
Table 14: Level of implementation of mWater by province ..... Art. 26
Table 15: Critical data from RPSPEs 27
Table 16:SWOT Analysis of PESRs ..... Art. 28
Table 17: Critical Data for EPSRA ..... Art. 35
Table 18:Inventory of existing IMS ..... Art. 37
Table 19: Status of data availability for AMAP indicators in existing IMS ..... Art. 38
Table 20: Proposed Governance Framework for IMS in the Water and Sanitation Sector sanitation in the DRC..... Art. 38
Table 21: Critical data domains to be standardized among managers and actors of the WASH sector ..... 39 *Table 22: Indicators collected in the NISWSS* 42
Table 23: Approach to integrating good practices into design and implementation of the SGI-EHA in the DRC ..... 47 *Table 24: Framework for Selecting the IMS Option* ..... 48 *Table 25: Overview of the flow between microservices, kernel, and IMS connections external* 50

Figure 1: Overview of the methodological approach..... 6
Figure 2: Overview of the RUWASA SDMS 44
Figure 3: Overview of the unified IMS-WASH 49

INTRODUCTION

1. General context of the PASEA project

The Government of the Democratic Republic of Congo has received support from the World Bank Group's International Development Association (IDA) to implement the Access to Water and Sanitation Services in the DRC (PASEA). The PASEA aims to increase access to basic drinking water supply and sanitation services in certain provinces of the DRC. The project focuses on the development of water and sanitation infrastructure with an overall objective of providing improved access to 12 million people for drinking water and 8 million for sanitation.

This program will be implemented according to the multi-phase programmatic approach, the first phase of which concerns the peri-urban and rural areas of the provinces of Kwilu, Kasai, Kasai Central and Kasai Oriental. Phase 2, whose approval will depend on the progress made under Phase 1, aims to expand the intervention to other ETDs among 5 other provinces (Ituri, Kongo Central, Lomami, North Kivu and South Kivu). Phases 3 and 4 will focus on sustaining service delivery, encouraging improved performance by provinces and service providers in all nine provinces.

The AMAP has four components:

1. *Improving Access and Capacity to Provide Drinking Water Supply Services*
   - Improving Access to Drinking Water in Rural and Peri-Urban Areas;
   - Improving the Performance of Private and Non-Profit Water Operators;
   - Strengthening Public Institutions and Capacities for Drinking Water Services
2. *Improving Access and Capacity for Sanitation Service Delivery*
   - Improving Access to Sanitation and Hygiene in Rural and Peri-Urban Areas;
   - Improving Access to Water, Sanitation and Hygiene (WASH) in Institutions;
   - Private Sector Development for Sanitation and Hygiene;
   - Strengthening Public Institutions and Capacities for Sanitation Services.
3. *Project Management, Learning and Scaling* ● Project Management and Learning; ● Phase 2 scaling.
4. *Conditional Emergency Response Mechanism*

This program reinforces the country's discussions for the gradual establishment of a real National Water-Hygiene-Sanitation (WASH) Platform with the aim of actively coordinating the WASH sector, monitoring and directing investments and optimizing transaction costs. It is expected that the WASH Platform will be operational by the end of 2024 or early 2025 at the latest. The Platform will have an important role to play in the collection of information related to the water and sanitation sectors. It therefore seems essential that the Task Force in charge of its implementation is well involved in the development of the Information Management Systems (IMS) that will be implemented through the Project. It is therefore necessary to diagnose the existing SGIs for both the water and sanitation sectors and to prepare the Terms of Reference to enable the Project to recruit one (or two) firm(s) that will develop one or two SGIs adapted for the drinking water sector and for the liquid sanitation and faecal sludge sector.

2. Objectives

The overall objective of the SGI consultant's mission is to develop detailed Terms of Reference after an in-depth diagnosis of existing SGIs and international best practices for water and sanitation information management.

Specific objectives

1. *Diagnosis of existing systems* : Analyze the SGIs currently in place in the water and sanitation sector in the DRC and other countries in the region (e.g. Zambia), identify gaps and needs.
2. *Stakeholder coordination* : Identify the actors involved in the collection, management and use of data (CEP-O, ARSPE, Provincial Boards, municipalities).

3. *Analysis of international experiences* : Present regional and international good practices, particularly in SGI WASH, with a particular focus on lessons learned from Zambia and other neighbouring countries.
4. *ToR Design* : Develop ToRs for the recruitment of firms specializing in the design of SGIs for water and sanitation

3. Methodological approach

To conduct the mission, the consultant relies on a structured approach, rooted in the joint use of the MERISE and C4Model methodologies, orients this mission to ensure tangible and measurable results

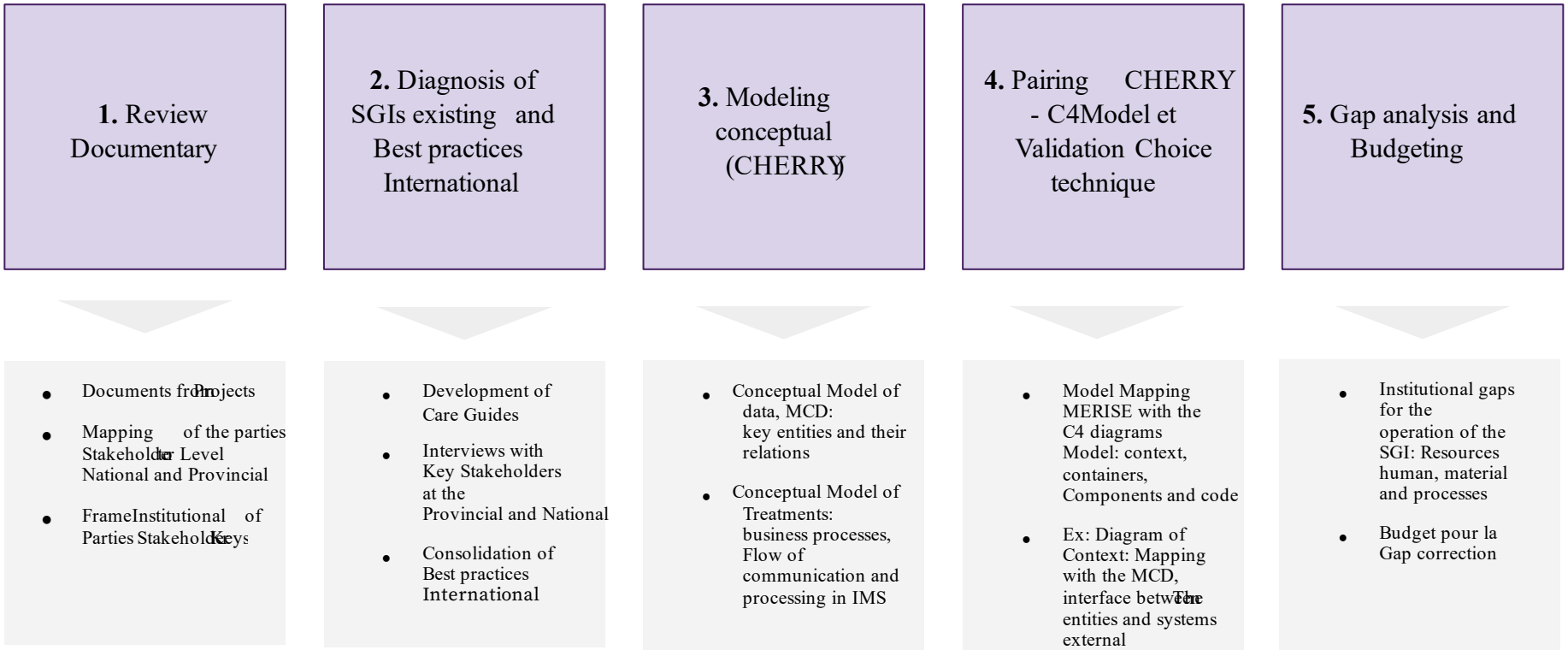

Figure 1 : Overvieof the Methodological

CHAPTER 1: DIAGNOSIS OF EXISTING SYSTEMS IN THE PROVINCES TARGETED

In this chapter, we present the diagnosis of the existing information management systems in the water and sanitation sector in the DRC as well as the needs expressed for their development.
The diagnosis was made in the organizations and public services involved in PASEA at the national level and in the provinces of Kwilu, Kasai, Kasai-Central and Kasai-Oriental.

National level	Provincial level
CNAEHA	Division Provincial de la sante
SAR	Provincial Water Utilities Board
DESP	ONHR Provincial Directorate
DHSP	CPEDD
DEVC	Hygiene Division
DINAC	DEVC Branch Coordinator
DRE	Division EPST
ONHR	Regional Directorate, REGIDESO
SG RHE	Governor
CSPP	Water Resources and Energy Division
REGIDESO	CPAEHA (CENAEHA Province)
	DINAC
	Public Procurement and Management Unit
	Provincial Minister of Budget and Finance

Table 2: Organizations covered by the diagnosis

The results of the diagnosis here provide an in-depth reading of the essential stakeholders in information management for the water and sanitation sector. The diagnosis made for the regulatory authority for public water services was made on the basis of the regulatory texts relating to its creation as not yet operational in the country.

1. Mapping of key stakeholders and status of existing IMS in structures

The water and sanitation sectors require coordinated actions by several ministries, each through its normative directorates and specialized structures for the planning, implementation and monitoring of the performance of strategic plans in the water and sanitation sector.

The reading of the water law of number 15/026 of 31 December 2015 and the national sanitation strategy allowed us to establish the mapping of the actors and their roles for a diagnosis of information management: the types of data produced or of interest, the mode of collection, management and sharing.

1. National Water, Hygiene and Sanitation Action Committee, CNAEHA

The National Water, Hygiene and Sanitation Action Committee (CNAEHA) is a multisectoral national body for strategic steering and coordination of actions in the water, hygiene and sanitation sector in the DRC. Created by decree to respond to the need for integrated and coherent governance, the CNAEHA brings together representatives of key ministries (Water Resources, Health, Education, Environment, Finance, etc.), public agencies, technical and financial partners, as well as civil society. It is placed under the direct supervision of the Prime Minister's Office, thus guaranteeing its political authority and its transversal role in the implementation of national WASH policies.

Its main missions include: (i) the development and monitoring of the national WASH policy, (ii) the coordination of sectoral plans and programmes, (iii) the harmonization of partners' interventions, (iv) support for resource mobilization, and (v) the supervision of governance, monitoring and evaluation and accountability mechanisms. In line with Articles 12 to 14 of Law No. 15/026 on Water, the CNAEHA acts as an advisory and consultation body to promote integrated, participatory and sustainable management of water resources. It also plays a fundamental role in facilitating sectoral dialogue and preventing water-related conflicts, in accordance with article 14 of the said law.

The CNAEHA is structured around a Strategic Steering Committee chaired by the Prime Minister, which brings together sectoral ministries, provincial representatives, technical partners and civil society, with the mandate to define the political orientations of the WASH sector, validate national plans and arbitrate institutional conflicts. Its operational functioning is ensured by a permanent Technical Secretariat, to which are attached several thematic technical commissions (drinking water, sanitation, WASH in schools and health structures, financing, monitoring and evaluation), responsible for technical analysis, intersectoral coordination and the development of normative tools, thus ensuring integrated and participatory governance in accordance with the requirements of the Water Law.

To carry out its mission, the CNAEHA has launched several initiatives for the implementation of an information management system in the water and sanitation sector: STUDY and SINIEau. These systems aimed to have an integrated and centralized database of all information in the water and sanitation sector for the development of policies and programmes, monitoring of implementation, monitoring of access and quality of services indicators as well as resource mobilization.

- **STUDY**

The STUDY initiative implemented as part of the support of the water program in urban and peri-rural areas aimed to support CANEHA in the mapping of water points in the country. This programme has made it possible to collect data and consolidate them in the form of a geospatial database accessible to the members of the technical commissions for exploitation and analysis. The backup of the database in CD format was shared with key players in the water sector.
The updating of the database required ad hoc data collection and was not included in the business processes of the structures providing the data in the provinces. Thus, the cessation of technical and financial assistance from the programme led to the cessation of data collection. The weak technical capacity of the CNAEHA did not allow for continuity in the management of the database.
At the time of the diagnosis of the information management systems for this report, the server was non-existent and no data backup database was available.

- **SINIEau**

The national integrated water information system, SINIEau, was supported by GIZ in 2011. The objective of the support was to support the Government of the DRC in drawing up the main lines of design for the implementation of the SINIEau. The SINIEau should make it possible to structure and strengthen the collection, treatment, storage and analysis of information in the water sector at the national, provincial and local levels.
Data should be stored in a unified, centralized database with user-friendly accessible interfaces, web interface with mapping capabilities, at all levels. The SINIEau should be managed by the CNAEHA.

The system included the management of data on the identification and classification of water sector actors, areas of operations (location), daily water production and projects in the water sector.
GIZ's strategic change led to the discontinuation of the initiative in its design phase.

The evolution of two past initiatives highlights the challenge of sustainability in the design and deployment of information management systems within the CNAEHA. Any future initiative should incorporate a management model into its design phase that promotes its sustainability.

- **Current state of information management**

The information management system (IMS) within the CNAEHA remains limited due to the failure of previous attempts to establish a centralized and integrated database for the water, hygiene and sanitation sector. In the absence of a common platform, each institutional actor collects and manages its own data according to its own methods, often without standardization or interoperability. Information is mainly shared on an ad hoc basis, through written reports or Excel files sent on request. The CNAEHA accesses the data by ad hoc requests addressed to structures such as the National Office of Rural Hydraulics (ONHR), the REGIDESO, the Provincial Boards, or the Directorate of Sanitation, which transmit the information in the form of Word memos or Excel tables. These data are then used by the functional commissions for the preparation of sectoral reports. However, there is no formal directive governing the management, retention or archiving of this data: each analyst keeps the files according to his or her own practices, which compromises the traceability, durability and security of the information.

Aware of the importance of a robust information management system for the achievement of its mission, the CNAEHA began discussions with partners at the end of 2024 for the revival of the implementation of an integrated and centralized database in the water, hygiene and sanitation sector.
The relaunch of this initiative has enabled the CNAEHA to clarify its data needs to inform its indicators of interest.
Critical data are summarized in the table below

Type	Description	Indicators	Collection Frequency
Demographics	Population by administrative level	Water access rate [disaggregated by administrative level]	Annual
Water points (boreholes, wells and springs)	Number, types, GPS location, operating status, quantity produced	Water access rate [disaggregated by administrative level]	Monthly
Water quality	Results of physico-chemical and bacteriological analysis	% of samples within quality standards	Daily / Monthly
WASH infrastructure (household, school and health care facility)	Households with access to improved latrines, Number of schools/health centres with access to water, latrines and washing facilities hands	% of households with access to improved latrines Rate of access to sanitation services	Daily / Monthly
Actors and areas of intervention	Directory of actors (type, identification, management method, etc.) and coverage area	Coverage gap	Monthly

Funding and ongoing projects	Amounts mobilised, status of projects, donors involved	Mapping of priority areas in	Quarterly
		Funding [disaggregated by administrative level] Gap in financing [disaggregated by level Administrative] % of projects at risk in implementation	
Governance and accountability	Steering committee meeting, committee working session	% of meetings held per year	Quarterly

Table 3: Critical dataCNAEHA

The analysis of strengths, weaknesses, opportunities and areas allows us to see the level of readiness of the CNAEHA for the implementation of a new IMS and to guide the design of a robust and perennial system.

Strengths	Weaknesses	Opportunities	Threats
Existence of a political and partnership will to relaunch the SGI	Lack of an integrated, operational platform	Existence of previous projects (STUDY, SINIEau) as a basis for learning	Multiple actors and dispersion of data without standardization mechanism
Institutional recognition of the CNAEHA by decree	Weak technical capacity to manage a sustainable IMS	Potential partner support for the revival of IMS (CEP-O, UNICEF, etc.)	Risks of stopping initiatives with the withdrawal of external funding
	Lack of IT equipment for IMS implementation	IT equipment support provided under the PASEA	

Table 4: SWOT Analysis of IMS at NAFHAA

- **Gap analysis in the implementation of the IMS within the CNAEHA**

The relaunch and operationalization of a functional information management system (IMS) within the CNAEHA requires the filling of several critical gaps, including human resources, physical resources and organizational processes.

Human resources : The analysis of the profiles of the staff of the Technical Secretariat of the CNAEHA reveals a lack of specific technical skills in database management, server maintenance, geographic information systems (GIS) and data analysis. This deficit, which had already been identified during the attempt to implement the STUDY system, had largely contributed to the abandonment of the initiative. To avoid a recurrence of this situation, it is imperative to build the capacity of existing staff rather than recruit new staff, in order to avoid the constraints of integration into the civil service and salary support. The capacity building program should be structured around the competencies essential to the administration and sustainability of IMS.

Physical Resources : The CNAEHA does not currently have adequate physical infrastructure to host or operate an IMS. The lack of a dedicated server, combined with an insufficient number of computers suitable for users, is a major obstacle. It is therefore necessary to provide for the acquisition of a minimum set of IT equipment including a secure server, workstations for key users, as well as connectivity and backup tools.

Processes and guidelines : The sustainable operation of an IMS also requires formalized procedures. Currently, there are no clear guidelines for the collection, consolidation, processing or dissemination of data. The development of standardized protocols is essential to ensure

harmonization of methods, compatibility of data between actors, as well as interoperability of systems at the national, provincial and local levels.

2. Sanitation Directorate, DAS

The Directorate of Sanitation (DAS), under the Ministry of Environment and Sustainable Development (MEDD), is the national technical body of reference for the governance of sanitation in the DRC. It develops and implements the national sanitation policy, sets technical standards and guidelines, supports provinces and territorial entities, and coordinates interventions with sectoral partners.

The DAS plays a key role in the implementation of the Open Defecation Elimination (ODF) Roadmap, providing response planning, monitoring provincial commitments, and supporting communities in adopting community-based approaches such as CLTS.
The DAS has four main technical functions: liquid waste and excreta management, solid waste and related structures management, vector control, and the analysis and control laboratory.
Liquid Waste and Excreta Management: The DAS oversees the planning, regulation, and monitoring of excreta management systems, including latrines, septic tanks, and faecal sludge treatment facilities. It develops technical standards relating to the construction, maintenance and emptying of autonomous and collective sanitation structures. *Management of solid waste and related works:* it defines technical guidelines for the management of household waste, sorting, collection, transport and treatment, as well as for the design and supervision of landfills, controlled dumps, and related structures (composting areas, storage pits). It also promotes sustainable community waste management practices.
Vector control: the DAS is involved in the planning and implementation of actions to combat vectors of environmentally-related diseases (malaria, filariasis, etc.), through the management of stagnant water, high-risk waste, and environmental disinfection in high-risk areas.
Analysis and control laboratory: the management has or collaborates with analysis laboratories specialising in the analysis of wastewater, excreta and solid waste, in order to ensure the sanitary quality of the structures and the compliance of discharges. It also supports quality control actions in urban centres and public institutions.

The DAS produces data throughout the performance of its four technical functions. To manage them, an information management system is necessary. The DAS, with the technical and financial support of UNICEF, has integrated DHIS2 as part of the Healthy Villages and Schools programmes.

- **District Health Information System 2, DHIS2**

DHIS2 is an open source health data management platform originally developed by the University of Oslo. Designed to be modular, extensible and customizable, this system is now used by more than 70 countries for the collection, analysis, visualization and monitoring of health data. In the DRC, DHIS2 is the health information management system in the DRC. It is maintained by the Directorate of the National Health Information System, DSNIS.
It was adopted by the DAS on the proposal of UNICEF as part of the Healthy Schools and Villages programmes.
A DHIS2 instance was set up to collect and aggregate sanitation data: latrine availability, access to water, handwashing facilities, and solid waste management in schools and villages.
DHIS2 has provided access to structural data to technical and financial partners. The

DHIS2 was discontinued as an information management system within DAS with the cessation of the Healthy Schools and Villages programmes. The cessation of its programmes has led to UNICEF taking over the accommodation.

- **Current state of information management**

The current information management system within the ASD is based on Excel files and paper forms. The Excel database is limited to data for monitoring the implementation of the ODF roadmap. Excel files are sent via email by the provincial coordination teams of the ODF roadmap. This mode of transmission does not guarantee the integrity, centralization and integration of data. The other technical functions of the DAS do not benefit from data management in a structured way, resulting in the loss of data and poor information sharing within the CNAEHA.

The DAS needs to rethink its IMS to ensure good data management. This IMS will need to integrate data on the four technical functions and the monitoring of the ODF roadmap.

In line with the new vision, the SGIs will integrate the following data:

Type	Description	Indicators	Collection Frequency
Demographics	Population by administrative level	Rate of access to Sanitation Services [disaggregated by administrative level]	Annual
FDAL Roadmap: Pre-Triggering	Funder, Roadmap Implementing Partner, Pre-Trigger Date, Start Time, End Time, Number of Existing Schools, Number of Schools with Latrines, Number of Existing Churches, Number of Existing Churches with Latrines, Other Existing Institutions with Latrines	Household sanitation access rate: % of households with latrines improved Rate of access to sanitation in schools: % of schools with functional latrines and washing facilities hands Rate of access to sanitation in schools: % of health centres with adequate sanitation facilities	Monthly
EDF Roadmap: Triggering	Date of triggering, Start time, End time, completion of the walk of shame, installation of an assistance committee, elaboration of an action plan, Number of households in the village, Number of households with latrine, Number of households with handwashing device, Preparation of the talking map, Number of men, Number of women, Number of girls under 18 years of age, Number of boys under 18 years of	Number of villages with the Validated OLF Household sanitation access rate: % of households with latrines improved Rate of access to sanitation in schools: % of schools with functional latrines and washing facilities hands	Monthly

	age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of boys under 18 years of age, Number of girls under 18 years of age, Number of girls under 18 years of age, Number of girls under 18 years of age, Number of	Rate of access to sanitation in	

Type	Description	Indicators	Collection Frequency
	of boys under 5 years of age, number of girls under 5 years of age, Sensitization on ODF	Schools: % of health centres with adequate sanitation facilities	
FDAL Roadmap: Post-release follow-up	Number of latrines built by households, number of latrines with handwashing facilities, number of households existing in the community, number of households with garbage holes, number of latrines in institutions and public places, number of latrines in institutions and public places with handwashing facilities, holding of the sanitation committee meeting, Visit of the members of the Committee of Community Facilitators	Household sanitation access rate: % of households with latrines improved Rate of access to sanitation in schools: % of schools with functional latrines and washing facilities hands Rate of access to sanitation in schools: % of health centres with adequate sanitation facilities	Monthly
FDAL Roadmap: FDAL Certification	Visit to the certification community, date of the certification visit, Certification Status ALF	Number of villages certifies FDAL Number of households living in the ADF-certified villages	Monthly

Faecal sludge management in urban and peri-urban areas	Name faecal sludge collection and treatment site, type, GPS location, site coverage area, site assessment according to standards, collection capacity and site treatment	Number of faecal sludge collection and treatment sites [disaggregated by level Administrative] Faecal sludge collection and treatment site treatment capacity [disaggregated by administrative level]	Monthly
Actor in faecal sludge management in urban and peri-urban areas	Identification of actors, coverage areas, quantity of faecal sludge supplied to the collection site and treatment of faecal sludge	Amount of faecal sludge produced [disaggregated by administrative level]	Monthly
Liquid waste and excreta management infrastructure in households and public spaces in urban and peri-urban areas	Location, type (septic tank, sewers, etc.), classification of the condition according to standards	Number of Liquid waste management infrastructure in standards	Quarterly
Solid waste management in urban and peri-urban areas	Name solid waste collection and treatment site, type (collection, treatment, collection and treatment), GPS location, Site coverage area, site assessment according to standards, collection capacity and site treatment	Number of solid waste collection and treatment sites [disaggregated by level Administrative] Treatment capacity of waste collection and treatment sites	Monthly
Type	**Description**	**Indicators**	**Collection Frequency**
		Solid [disaggregated by administrative level]	
Actor in solid waste management in urban and peri-urban areas	Identification of actors, coverage areas, quantity of solid waste supplied to the collection site and treatment at the solid waste collection and treatment site	Quantity of waste supplied to the solid waste collection or treatment site [disaggregated by administrative level]	Monthly
Funding and ongoing projects	Amounts mobilised, status of projects, donors involved	Mapping of Priority Areas for Funding [disaggregated by administrative level] Gap in financing [disaggregated by level Administrative] % of projects at risk in implementation	Quarterly
Governance and accountability	Steering committee meeting, committee working session	% of meetings held per year	Quarterly

Table 5:SAR critical data

The prevention of data to be collected by the IMS at the DAS level demonstrates great ambition to inform decision-making, monitoring and evaluation of ongoing initiatives within the framework of the national water, hygiene and sanitation plan as well as strategic planning.

However, a question arises about the level of preparations of the DAS for the implementation and sustainability of such an IMS.

Strengths	Weaknesses	Opportunities	Threats
Clear institutional mandate for sanitation at the national level	Past dependence on shelter and external funding (UNICEF)	Willingness to relaunch an integrated IMS covering the four technical functions	Risk of loss of non-centralized or unbacked data
Previous experience with DHIS2 to collect data in programs Healthy schools/villages	Data included on DHIS2 is related to the objectives of the Healthy Schools/Villages program and not embedded in the technical functions of the DAS	Historic partnership with donors (UNICEF, Global Fund, etc.) for technical and Financial	Data fragmentation and manual transmission via files Unsecured
	Current lack of a functional centralized platform for all technical functions	Provision of computer equipment under the PASEA	

Table 6: SWOT Analysis of SAR

- **Gap analysis in the implementation of IMS within the DAS**

To ensure optimal and effective implementation of the Information Management System (IMS) within the Directorate of Sanitation (DAS), several resource constraints must be overcome.
Human resources : The analysis of staff profiles shows a lack of technical skills in key areas such as database management, server maintenance, geographic information systems (GIS) and data analysis. This deficit is an obstacle to the sustainable ownership of IMS and calls for targeted internal capacity building.
Physical resources : DAS does not currently have adequate IT infrastructure to host or operate an IMS. However, an allocation of basic equipment is planned under the PASEA programme, which represents a concrete opportunity to be seized to fill this gap.
Processes and Guidelines : Relevant data—as detailed in Table 5—are collected at different levels of government (province, territory, sector/chieftaincy, neighbourhood/grouping, village/avenue). However, the lack of standardized collection tools between these levels limits the integration and consistency of information. The production and validation of harmonised and standardised tools is therefore essential to ensure better interoperability of data between administrative levels and with other information systems (such as those of the CNAEHA or the DHSP).

3. National Office of Rural Hydraulics, ONHR

The National Office of Rural Hydraulics, ONHR is a public technical institution under the supervision of the Ministry of Hydraulic Resources and Electricity (MRHE). Its main mission is to ensure the planning, construction, monitoring and maintenance of drinking water supply infrastructure in rural and peri-urban areas of the DRC. Its role is central to the implementation of the national water policy, particularly with regard to improving equitable and sustainable access to drinking water for populations living outside urban centres.

ONHR is involved in all stages of the rural water project cycle: feasibility studies, resource mobilization, supervision of works, support for the establishment of local water management committees, and capacity building of local entities. It collaborates with decentralized territorial entities (ETDs), NGOs, technical and financial partners, and other public operators such as REGIDESO or provincial authorities, as part of a multisectoral approach to water, hygiene and sanitation (WASH).

ONHR has a central level based in Kinshasa, which ensures national technical coordination, the development of standards, the supervision of work, and the capitalization of data. This central level is structured into technical units according to the key themes: hydrogeological studies, quality control, monitoring and evaluation, management of standards and database.
At the provincial level, ONHR is represented through provincial directorates. These directorates are responsible for supervising the work carried out locally as part of water supply projects, ensuring the quality control of structures, providing data at the central level, supervising drilling or maintenance teams, and supporting ETDs in the planning of their rural hydraulic needs.
A vertical communication and reporting circuit exists between the provincial directorates and the central directorate. The provincial directorates produce monthly technical reports, supported by standardized sheets (paper or Excel), which are sent to the central level for consolidation and operation. However, the lack of an integrated computerized system often hinders the smooth transmission and traceability of data.

As the technical arm of the Ministry in rural areas, ONHR is also a key player in the production and management of rural water data. It participates in the monitoring and evaluation efforts of national programmes such as the PNEHA and PASEA, providing data on functional water points, the types of technologies used, the quality of structures and rehabilitation needs. It plays an important role in the sector's information management system (IMS), although its data collection and archiving capabilities still need to be strengthened.

ONHR is therefore positioning itself as a strategic player in achieving the objectives of universal access to drinking water in the DRC, particularly in areas where traditional public services are absent or inefficient. As such, he is regularly called upon in sectoral coordination frameworks, such as the CNAEHA and the steering committees of WASH projects, for his knowledge of the field and his technical expertise.

- **Current situation of information management within ONHR**

Currently, data management within the National Office of Rural Hydraulics (ONHR) is essentially based on non-centralized and insecure media. Technical information is kept in the form of paper reports or digital documents (Word or PDF), while databases are kept in Excel format. However, these files are not hosted on a shared or structured storage space: they remain locally on the individual computers of the staff members. This situation exposes ONHR to a high risk of data loss, redundancy of information, and severely limits efficiency in the search for and use of data for analysis, planning or monitoring and evaluation purposes.

Provinces	Information Management Method	Comment
Kwilu	Paper and Word documents	No archiving system
Kasai	Paper and Word documents	No archiving system

Kasaï-Central	Database in Excel file format and support for paper technical reports	No archiving system and data hosted on individual staff computers
Kasaï-Oriental	Database in Excel file format and support for paper technical reports	No archiving system and data hosted on individual staff computers

Table 7: Existing IMS within the ONHR provincial directorates

In 20218, ONHR produced a framework of work standards in the water sector also highlighting the types of data useful in the management of water point databases.

Type	Description	Indicators	Collection Frequency
Demographics	Population by administrative level	Water access rate [disaggregated by administrative level]	Annual
Water points (boreholes, wells and springs)	Number, types, GPS location, study report for its implementation, technical characteristics of the structure, operating condition, water quality certification report during implementation	Number of functional water points [disaggregated by administrative level]	Monthly
Water quality	Results of physico-chemical and bacteriological analysis	Number of water points in the quality standards at commissioning [disaggregated by administrative level and by users' association or local committee water]	Daily / Monthly
Local Water Users Association or Committees	Directory of user associations, water points under their	Quantity produced by the water point [production, reservoir, distribution]	Daily / Monthly
	management, daily production per water point, daily revenues, water sales tariff at the water point, maintenance intervention at the water point and their costs	Average access time day in the water Recipe from the sale of water [revenue and applied price] % of non-billed water at the water point [index	
Funding and ongoing projects	Status of mobilization of resources for the realization of studies, status of studies, donors involved, levels of achievement as soon as the implementation of water point projects	Number of studies of non-funded water points [disaggregated by level Administrative] Gap in financing [disaggregated by level Administrative] % of projects at risk in implementation	Quarterly

Table 8: Critical data from ONHR

Although ONHR has identified a set of critical data for monitoring its activities, it is struggling to be integrated into a real information management system (IMS). To operationalize this data framework effectively, it is necessary to remove certain technical, organizational and structural constraints. To this end, the analysis below presents the strengths, weaknesses, opportunities and threats (SWOT) that ONHR faces in the implementation of a functional IMS.

Strengths	Weaknesses	Opportunities	Threats
Clear institutional mandate in the management of rural water	Lack of a centralized platform for archiving and sharing data	Possibility of integration into the sectoral IMS under development (PASEA, CNAEHA)	High risk of loss of critical data in the event of a breakdown or staff turnover
Existence of regularly generated technical data on the Terrain	Data stored locally on staff computers without secure backup	Interest crescent Partners for harmonization WASH data	Gap between field practices and the standards expected in modern IMS
Presence at provincial offices for local data collection	Lack of qualified database management personnel and Information Systems	Possible technical support for digitalization and staff training	Data fragmentation limiting consolidation and analysis at the central level

Table 9: SWOT analysis of the ONHRS

- **Analysis of gaps in the implementation of the IMS within ONHR**

The implementation of an effective information management system (IMS) within ONHR requires filling several organizational, technical and structural deficiencies that currently hinder the optimal use of data.

Human resources : ONHR staff, both at the central and provincial levels, have little specialization in areas related to database management, server administration, geospatial database structuring and data analysis. This lack of technical skills is a major obstacle to the operationalization of a sustainable IMS. Targeted capacity-building actions need to be considered, including in the areas of hydraulic data management, digital reporting, data quality control and the use of GIS software.

Material resources : currently, ONHR has neither a dedicated server to host its databases, nor a shared platform to centralize information from provincial branches. Data is stored locally on agents' computers, increasing the risk of loss, duplication and breaking in the information chain. The provision of a minimum of IT equipment (centralized server, secure computers, backup solutions) is essential to ensure the traceability and durability of the data collected.

Process and guidelines : Although ONHR regularly collects data on rural water works, the collection and transmission tools vary greatly from one province to another. There are no standardised forms or harmonised processes for reporting, archiving and using data. In addition, the integration of these data into sector planning mechanisms remains limited. The development and adoption of standardized operating procedures (SOPs) and harmonized frameworks are required to ensure consistency, quality and compatibility of data at the national level, in conjunction with existing systems such as those of the CNAEHA or programs supported by the ASEAP.

4. Hygiene and Public Health Directorate, DHSP

The Directorate of Hygiene and Public Sanitation (DHSP) is a central technical structure of the Ministry of Public Health, Hygiene and Prevention (MSPHP). Its mission is to ensure the protection of public health through the improvement of hygiene and sanitation conditions in all living environments: households, schools, healthcare facilities, public places, and professional

environments. It develops policies, strategies, standards and guidelines related to environmental hygiene, domestic sanitation, biomedical waste management and sanitation of sanitation infrastructure.

The DHSP plays a transversal role in the Water-Hygiene-Sanitation (WASH) sector, in particular for the sanitary control of water quality, the management of hygiene in health structures, and the support of communities in the adoption of good hygiene practices in coordination with the Directorate of Care Establishments and Partnerships (DESP), the Hygiene and Public Health Offices (BHSP) at the provincial level, health zones, as well as with technical partners (UNICEF, WHO, etc.). The DHSP is involved with the DAS in the implementation of the ODF roadmap.

The DHSP is organized into specialized technical units at the central level, each responsible for themes such as domestic hygiene, institutional sanitation, water health control, or medical waste management. For implementation, it relies on the BHSPs (Hygiene and Public Health Offices) established in each province, which are the operational relays of the MSPHP. These offices coordinate hygiene actions in health zones and ensure the collection of health data related to sanitation.
Activity reports, supervision sheets, health control results are transmitted from the BHSP to the DHSP via a vertical reporting circuit. These data feed into the national dashboards and the meetings of the National WASH Action Committee (CNAEHA) in which the DHSP actively participates.

- **Current state of information management**

The DHSP operates the District Health Information System 2 (DHIS2), the main information management system for the health sector in the DRC. Through this system, the DHSP monitors several indicators related to hygiene in health facilities, medical waste management, and water health compliance. However, data on community hygiene and environmental health outside of health care settings are often reported in paper or Excel format, without full integration into a unified sector IMS.

This situation limits the full visibility of hygiene interventions in communities and hinders multisectoral analysis with other actors in the WASH sector (CNAEHA, DAS, ONHR, etc.).

To improve information management, the DHSP initiated technical discussions at the end of 2024 for the integration of a digital data collection tool. These discussions, although in the initial phase, have mapped the data needs in the WASH sector.

Type	Description	Indicators	Collection Frequency
Demographics	Population by administrative level	Rate of access to Sanitation Services [disaggregated by administrative level]	Annual

FDAL Roadmap: Pre-Triggering	Funder, Roadmap Implementing Partner, Pre-Trigger Date, Start Time, End Time, Number of Existing Schools, Number of Schools with Latrines, Number of Existing Churches, Number of Existing Churches with Latrines, Other Existing Institutions with Latrines	Household sanitation access rate: % of households with latrines improved Rate of access to sanitation in schools: % of schools with functional latrines and washing facilities hands Rate of access to sanitation in schools: % of health centres with adequate sanitation facilities	Monthly
EDF Roadmap: Triggering	Date of triggering, Start time, End time, completion of the walk of shame, installation of an assistance committee, elaboration of an action plan, Number of households in the village, Number of households with latrine, Number of households with handwashing device, Preparation of the talking map, number of men, number of women, number of girls under 18 years of age, Number of boys under 18 years of age, Number of boys under 5 years of age, number of girls under 5 years old, Awareness raising on ODF	Number of villages with the Validated OLF Household sanitation access rate: % of households with latrines improved Rate of access to sanitation in schools: % of schools with functional latrines and washing facilities hands Rate of access to sanitation in schools: % of health centres with adequate sanitation facilities	Monthly
FDAL Roadmap: Post-release follow-up	Number of latrines built by households, number of latrines with handwashing facilities, number of households in the community, number of households with garbage holes,	Household sanitation access rate: % of households with latrines improved Rate of access to sanitation in schools: % of schools with	Monthly
Type	**Description**	**Indicators**	**Collection Frequency**
	Number of latrines in institutions and public places, number of latrines in institutions and public places with handwashing facilities, holding of the meeting of the sanitation committee, visit of the members of the committee of community facilitators	functional latrines and washing facilities hands Rate of access to sanitation in schools: % of health centres with adequate sanitation facilities	

FDAL Roadmap: FDAL Certification	Visit to the certification community, date of the certification visit, Certification Status ALF	Number of villages certifies FDAL Number of households living in the ADF-certified villages	Monthly
Inspection of classified establishments	Facility Identification, Facility Classification, Inspection Date, Remediation Inspection Note	Number of establishments classified in compliance with the standards of sanitation [functional latrines, access to water, handwashing facilities]	Quarterly
Health Care Establishment	Identification of the health facility, classification of the health care facility, Date of inspection, sanitation inspection note	Number of health facilities in compliance with sanitation [functional latrines, access to water, handwashing facilities] Number of health facilities with a biomedical waste management plan functional	Quarterly
Water quality	Results of physico-chemical and bacteriological analysis	Number of water points in the quality standards at commissioning [disaggregated by administrative level and by users' association or local committee water]	Daily / Monthly
Prevalence of waterborne diseases	Name of waterborne diseases, number of cases	Prevalence of waterborne diseases	Monthly

Table 10:Critical data from the DHSP

In order to identify the internal and external levers favorable to the implementation of an effective information management system within the Department of Hygiene and Public Sanitation (DHSP), a SWOT analysis was carried out. This analysis highlights the existing strengths, operational constraints, prospects for collaboration as well as the risks to be anticipated in the development of an IMS adapted to the functions of the DHSP.

Strengths	Weaknesses	Opportunities	Threats
Data Integration hygiene/health in DHIS2	Poor BHSP digital equipment	Possible interoperability with sector IMS (CNAEHA, PASEA)	Risk of information loss in areas not digitally covered
Attendance of Relay through BHSP	Lack of harmonized tools for community collection, water quality monitoring and inspection of classified facilities	Existing partnerships with WHO, UNICEF, and others TFP	Weak integration with environmental data (MEDD, DAS)

Table 11:SWOT analysis of the DHSP

- **Gap analysis for the implementation of SGI within the DHSP**

Human resources : DHSP managers are proficient in DHIS2 for health data, but capacity for expanded information systems, environmental database management, and interoperability remains to be strengthened, especially at the provincial level. *Physical resources* : Although the central level has basic IT capabilities, not all BHSPs are equipped to digitally capture or transmit data, resulting in the use of paper and long processing times.
Process and guidelines : There is no unified collection framework for all DHSP themes. The tools differ by province, and their compatibility with other IMS (such as those of the CNAEHA or the MEDD) is limited. Standardizing forms, automating reporting, and instituting sharing and saving guidelines are essential.

5. General Secretariat at the Ministry of Water Resources and Electricity, SG-MRHE

The General Secretariat at the Ministry of Water Resources and Electricity (SG-MRHE) is the central administrative and technical body responsible for the implementation of the Government's policies, strategies and programmes on water resources management and electricity production in the Democratic Republic of Congo. It ensures the coordination of technical directorates, the application of ministerial guidelines and the articulation of sectoral actions with the other ministries and institutions concerned.

In the field of water, the SG-MRHE plays a key role in supervising the public water service, regulating the exploitation of water resources, and monitoring the implementation of Law No. 15/026 on water. It is also the interface between the central and provincial levels to ensure integrated and equitable governance of access to drinking water, in particular through the monitoring of structuring programmes (REGIDESO, ONHR, RPSPE, PASEA, etc.).
The SG-MRHE has two technical directorates in the water sector: the Small Water Production Units Directorate and the Medium and Large Water Production Units Directorate

- **Directorate of Small Water Production Units and the Directorate of Medium-Sized Enterprises** This directorate is responsible for monitoring, planning and supervising small-scale water production systems, generally located in rural and peri-urban areas. These systems include motorized boreholes, mini-water supplies, developed springs or community systems managed by ETDs or user committees.

Its main missions include: the identification and mapping of small units in operation or planned, the technical validation of structures and the control of their compliance with the standards in force, technical support for rural water projects in conjunction with ONHR, NGOs and provincial authorities, the collection and centralization of performance data, The production and sustainability of these units, the management is a key player in the efforts of territorial equity and the reduction of inequalities in access to water.

- **Directorate of medium and large water production units**

This directorate provides technical supervision of medium- and large-scale water production infrastructure, mainly in urban areas and in high-growth semi-urban centres. In particular, it is involved in REGIDESO facilities, public collection and treatment plants, and major drinking water supply projects.
Its functions include: the technical evaluation of structuring projects in the field of urban water, monitoring the compliance of the technologies used with national standards, coordination with the Directorate of Urban Hydraulics, REGIDESO and donors, monitoring the performance of production, treatment and distribution on a large scale. This direction is essential to ensure the

quality, resilience and sustainability of urban water systems and their extension to unserved peripheries.

The two directorates play a key role in the allocation of research permits and the granting of hydraulic perimeters.
Granting of exploration permits: the technical analysis of the files submitted for the search for groundwater or surface water resources, whether for community, institutional or private uses, the issuance of a technical opinion prior to the issuance of the permit by the competent authority (the Minister or the Governor, as the case may be), the monitoring of hydrogeological studies and the validation of reports and the compliance of water points with sustainability requirements, and protection.
Allocation of hydraulic perimeters: the technical delimitation of production or catchment perimeters, in collaboration with the provincial services and local water committees, the proposal for the allocation of perimeters to public, private or community operators, according to the criteria of availability of the resource, population density and territorial planning, the transmission of allocation proposals to the Public Service Regulatory Authority (ARSPE), for registration and monitoring of the legal framework.

- **Current state of information management**

The General Secretariat of the MRHE, through its technical directorates in charge of water production, does not yet have a fully operational integrated information management system (IMS). The management of data relating to hydraulic structures, water points, permits and projects is still largely done manually or via non-centralised Excel files, often kept on the personal computers of technical managers. This situation limits the consistency, traceability and accessibility of the strategic information needed for planning and monitoring the sector.

However, under some programs supported by technical and financial partners, including the RESE project, PASEA, the SG-MRHE has started to use the mWater platform as a tool for collecting, visualizing, and managing data on water points. mWater is an open-source, cloud-based tool specifically designed for managing water, sanitation and hygiene services. It allows the collection of geolocated data via mobile, the real-time monitoring of the functionality of the structures, and the creation of interactive dashboards for analysis.
The adoption of mWater has enabled certain departments, in particular those in charge of small production units, to structure information on existing structures, their operating status, the technologies used, as well as the rehabilitation needs. Nevertheless, the use of mWater remains fragmented, project-driven, without being fully institutionalized or interoperable with other sectoral management systems such as those of CNAEHA, REGIDESO or ARSPE.

As currently used within the HEM-SG, mWater does not fully address the overall information management needs. Databases are fragmented and often limited to provincial water utilities, which prevents the consolidation of a unified national database. In addition, the platform does not integrate critical information related to the granting of research permits and the allocation of hydraulic perimeters. It is also not interoperable with other information management systems in the sector, such as those of the CNAEHA, ARSPE or DHIS2, limiting inter-institutional coordination.
That said, mWater remains a powerful tool for field data collection, offering a free online aggregation interface and offline input capabilities, making it an accessible and useful solution in low-connectivity settings. Its potential can be fully exploited if it is integrated into a clear institutional strategy that is articulated with the other sector SGIs.

This integration would be all the more effective if it was based on a comprehensive mapping of the critical data to be collected and monitored by the SG-MRHE. The table below presents the main types of data and key indicators corresponding to the missions of the technical directorates concerned.

Type	Description	Indicators	Collection Frequency
Demographics	Population by administrative level	Water access rate [disaggregated by administrative level]	Annual
Water points (boreholes, wells and springs)	Number, types, GPS location, operating status, quantity produced	Water access rate [disaggregated by administrative level]	Monthly
Medium and large water production units	Technical characteristics of the infrastructures (stations pumping, treatment) Volume of water produced and distributed (by city or urban centre)	Number and capacity of stations in operation Volume of water distributed daily / urban coverage rate	Monthly
Water quality	Results of physico-chemical and bacteriological analysis	% of samples within quality standards	Daily / Monthly
Actors and areas of intervention	Directory of actors (type, identification, management method, etc.) and coverage area	Coverage gap	Monthly
Granting of research permits	Mapping of the area covered, study deadline, study report	Number of permits issued by type and province	Monthly
Allocation of hydraulic perimeters	Perimeter mapping, perimeter assignment date, full study report and files	Number of Scopes Affected / Number Pending	Monthly
Pricing	Pricing	Number of Pricing Orders Promulgated on/Pending Orders	Semi-annual
Funding and ongoing projects	Amounts mobilised, status of projects, donors involved	Mapping of Funding Priority Areas [disaggregated by administrative level]	Quarterly
Type	**Description**	**Indicators**	**Collection Frequency**
		Gap in financing [disaggregated by level Administrative] % of projects at risk in implementation	

Table 12:Critical data from the SG-MRHE

The SWOT analysis highlights the existing assets, operational constraints, prospects for collaboration as well as the risks to be anticipated in the development of an IMS adapted to the functions of the SG-MRHE.

Strengths	Weaknesses	Opportunities	Threats
Clearly defined technical mandate for the management of water production at all scales	Fragmentation of data between provincial structures and the central level	Potential for integration with CNAEHA, ARSPE and platforms such as DHIS2	Risks of critical data loss due to insecure local storage
Presence of existing data via mWater and network networks Provincial	Lack of an integrated system for the management and analysis of permits and hydraulic perimeters	Technical and financial support available through structuring programs (e.g. PASEA)	Lack of standardization that can affect data quality and use

Table 13:SWOT Analysis of SG-MRHE

- **Gap analysis for the implementation of an IMS – SG-MRHE**

The implementation of an effective information management system (IMS) within the General Secretariat of the Ministry of Water Resources and Electricity (SG-MRHE) requires the removal of several major constraints related to human and material resources and management processes.
Human resources : The technical directorate teams have solid expertise in hydraulic engineering and monitoring water production projects, but they have significant gaps in data management. The specific skills required to administer databases, use geographic information systems (GIS), or ensure interoperability between sectoral platforms are limited, both at the central and provincial levels. Targeted technical capacity building in information management, data visualization, quality control, and interconnection between systems is a priority.
Material resources : The central level is relatively better equipped, but the provincial directorates do not have sufficient IT equipment or appropriate tools to ensure reliable data entry, rapid transmission or secure backup. The lack of a centralized server and shared storage space prevents the national consolidation of critical data, increasing the risk of information loss, duplication or breakage. It is therefore necessary to equip the relevant departments with basic equipment (servers, computers, backup and connectivity solutions).

Processes and Guidance : Data collection tools vary from province to province and are not harmonized or aligned with common standards. There is no standardized framework for the feedback, validation or archiving of technical data. This heterogeneity prevents a smooth integration with the information systems of other institutions in the sector, such as the CNAEHA, the ARSPE or the DHIS2. To address these shortcomings, it is imperative to develop standard operating procedures, validate common forms at the national level, and establish a clear protocol for managing and sharing information.

6. Provincial Water Utilities Board, PEPR

The Provincial Water Service Boards (RPSPE) are technical structures created by order of the provincial governments in accordance with Article 76 of Law No. 15/026 of 31 December 2015 on water. They constitute the operational arm of the province to exercise the project management of public water services, in a context of decentralization and mandatory delegation of the service to third-party operators (ASUREP, farmers, private delegates).

Their role is to steer the planning, contractualization, monitoring of delegated management and monitoring the performance of drinking water services in the provinces concerned. They are also involved in governance, accountability and resource mobilization in support of the provincial sector strategy.

The RPSPEs carry out their functions through the five dimensions of project management:
Planning: inventory of structures, needs assessment, development of financing plans
Resource mobilization: partnership development, funding management, monitoring of mobilization plans
Implementation of the works: supervision of studies (APS, APD), awarding of contracts, acceptance of the works
Monitoring of operations : control of the execution of delegation contracts, performance evaluation, conflict resolution
Governance and accountability : monitoring and evaluation of the service, accountability, facilitation of the provincial sectoral dialogue.

The RPSPEs are organized internally into three pools under the management of a coordinator.
Administrative and financial pool : budget management, HR, contracts, reporting.
Technical pool : monitoring of structures, planning, water quality, supervision of delegates. *Legal pool* : legal support, contract review, regulatory compliance.
This organisation allows for multidisciplinary and integrated management of services, in close collaboration with public service delegates

- **Current state of information management**

The information management system (IMS) within the Provincial Water Utilities Boards (PPWS) is currently based on the mWater platform, set up with the technical and financial support of GIZ. This solution was chosen for its flexibility, free accessibility, and mobile collection and online data aggregation capabilities, even when offline. mWater is now operational in several provinces, where it is used by utilities to structure the management of water service information.

In its deployment, each provincial authority has its own mWater database instance, managed autonomously. This decentralized architecture provides flexibility but does not allow for communication between provincial databases or automatic consolidation of data at the national level. In addition, the lack of a framework for standardizing collection tools and data entry methods has led to heterogeneous practices between provinces, which hinders the harmonization of indicators and the comparability of results.

The data collected and used on mWater mainly concern three dimensions: *Identification and mapping of water infrastructures* (water points, networks, stations,
etc.) ;
The registration of water sector operators, whether community, public or private;
Monitoring of the performance of delegated operators, through 14 key indicators, covering in particular service continuity, recovery rate, water quality, and user satisfaction.

Provinces[1]	**Implementation status of the mWater**	**Data available on the platform**
Kwilu	Effective mWater deployment	Georeferenced database of available water infrastructure and operator identification
Kasai	mWater Deployment Underway	NA
Kasaï-Central	mWater Deployment Underway	NA

[1] Focus on the provinces of implementation of Phase 1 of the PASEA

Kasaï-Oriental	Effective mWater deployment	Georeferenced database of water infrastructure, identification of available operators, monthly reports to monitor operator performance and water quality analysis database available

Table 14: Level of implementation of mWater by province

While mWater provides real value in data collection, visualization and analysis at the local level, its current use in the ESPPRs has several structural limitations. The fragmentation of databases by province, the lack of a nationally consolidated platform, the lack of interoperability with other sectoral systems (CNAEHA, ARSPE, SG-MRHE).

In addition, several critical data necessary for the complete management of the public water service are not yet integrated into the current system: data on contracts, finances, quality of service, local governance, or the monitoring of operators' contractual commitments. For IMS to fully play its role, it must become a real central working tool for the provincial authorities and all contracted operators, covering a wider spectrum of data and integrating hierarchical validation and approval mechanisms at different levels (local, provincial, central).

This evolution raises the need to expand the digital ecosystem beyond mWater, which remains primarily a collection and visualization tool, without offering the contract management, multi-source consolidation, and data governance capabilities expected from a robust IMS. It is therefore essential to design an integrated and scalable architecture, built around mWater for field collection, but enriched by a decision-making, analytical and operational core at the level of the networks and the sector.

The more complete mapping of critical data makes it possible to guide this integrated and evolving IMS architecture within the RPSPEs.

Type	Description	Indicators	Collection Frequency
Demographics	Population by administrative level, population served, number of connections, coverage area	% of water coverage	Monthly

Water infrastructure	Number, types, GPS location, operating status, operating hours per day, number of active subscribers, volume of water produced, volume of water distributed, network performance,	Water access rate [disaggregated by level Administrative] Hour of operation per day Evolution of Connections Network Performance Volume of non-billed water	Monthly
Financial Management	Cost of production per m3, selling price per m3, recovery rate	Water rates applied to m3 Operating margin	Monthly
Outages and complaint management	Outages, Outages Date, Description, Type, Handling Time, Escalation Date, Resolution Date, Complaint Date, Complaint Type, Complaint Description, Complaint Escalation Date, Complaint Resolution Date	% of complaints processed within the time limit Number of interruptions Services Average hours between Fault reporting and resolution	Monthly
Water quality	Results of physico-chemical and bacteriological analysis	% of samples within quality standards	Daily / Monthly
Contract management	Dossier of Contractualization, Operator Specifications	Operator Performance Score[2]	Monthly
Funding and ongoing projects	Amounts mobilised, status of projects, donors involved	Mapping of Funding Priority Areas [disaggregated by administrative level] Gap in financing [disaggregated by level Administrative] % of projects at risk in implementation	Quarterly

Table 15: Critical data from the PSPRs

The SWOT analysis allows us to assess the level of readiness of the RPSPEs for the adoption of an integrated and scalable IMS that takes into account all critical data. This analysis guides the design of the new IMS for optimal and perennial operation.

[2] Need to harmonize the weight of each indicator in the calculation of the composite score of operators' performance scores

Strengths	Weaknesses	Opportunities	Threats
Effective deployment of mWater in several provinces (Kwilu, KasaïOriental)	Fragmentation of mWater databases without interconnection between provinces	Possibility of expanding the digital ecosystem around mWater via Add-ons	Technical dependence on partners (e.g. GIZ) for the maintenance and evolution of mWater
Robust, accessible, mobile and Offline usable	Lack of a consolidated system for national data governance	Existing support from the PASEA project to structure a Harmonized SGI	Risk of increasing divergence of practices between provinces in Lack of standardization
Active use of mWater for mapping infrastructure, identification and performance monitoring	Failure to integrate critical data on contracts, finance, governance and commitments of Operators	Dynamic Growing institutional focus on performance and accountability of services)	Lack of interoperability with sector-specific IMS (CNAEHA, RHEMS, SPEAR), limiting analysis multi-sector

Table 16:SWOT Analysis of the SWOT of the PRSPEs

- **Gap analysis for the implementation of an IMS within the RPSPEs**

The transition to an institutionalized, interoperable and operational IMS in the Provincial Boards requires the removal of several technical, human and organizational constraints identified through current experiences of using mWater. *Human resources:* The utilities have operational staff experienced in the management of water services, but specific skills in information management, database administration, interoperability, and advanced data visualization remain limited. In addition, there is no unit dedicated to monitoring and analyzing data in several agencies, which hinders the regular use of the information collected via mWater. Targeted capacity building, particularly on the strategic use of data for management and contractual monitoring, is needed.
Material resources: the IT tools available vary greatly between agencies. While some have working hardware, the majority suffer from limited access to high-performance computers, stable connectivity, backup tools, or local servers.
The lack of a secure environment for data archiving and a centralized platform limits nationwide consolidation.
System and architecture: mWater, currently in use, works as a local collection and aggregation solution, but does not allow for centralized management or interconnection between bases in different provinces. The system does not cover some critical aspects such as: contract management, operators' financial data, local governance, and investment plans. The lack of an integrated analytical core prevents a consolidated view of performance at the national level and makes it difficult to make decisions based on multi-source data.
Normative framework and guidance: There are not yet standardized national or provincial guidelines for the use of IMS within the authorities. Collection and capture tools are customized to local preferences, resulting in inconsistencies, poor comparability of data, and loss of quality. A harmonized framework (standards, frameworks, validation guides and approval mechanisms) is essential to ensure the uniformity, reliability and traceability of data over time.
Governance and interoperability: the current system is poorly integrated with other sectoral platforms (SG-MRHE, ARSPE, CNAEHA), which prevents the implementation of integrated data governance in the WASH sector. The lack of an institutional strategy on information management at the national level reinforces this fragmentation. The development of an IMS articulated around mWater but enriched by a central decision-making layer represents a major avenue for structuring a sustainable digital ecosystem.

7. General Secretariat at the Ministry of National Education and New Citizenship

The Ministry of National Education and New Citizenship plays a cross-cutting role in improving access to water, hygiene and sanitation, particularly in schools. Through its technical directorates, the Ministry contributes to the design of adequate school infrastructure, the promotion of responsible civic behaviour, and the integration of hygiene education into school curricula. It thus acts as a central partner in the implementation of WASH policies, in conjunction with the Ministry of Health, the Ministry of the Environment, and the CNAEHA.

- **The National Directorate of Construction, DINAC**

The National Directorate of Construction, DINAC is responsible for the planning, monitoring and supervision of school infrastructure, including water, hygiene and sanitation facilities. It is involved in the definition of construction standards, the validation of architectural plans, and the coordination of school rehabilitation and modernization projects.
Its key missions are: development of technical standards for sanitary blocks, water points and hygiene equipment in schools, integration of WASH infrastructure into school infrastructure investment plans, technical supervision of the construction or rehabilitation of schools with WASH equipment, monitoring of projects financed by partners (UNICEF, World Bank, AFD, etc.) To ensure compliance with national standards, support to provincial inspections for feedback on the condition of infrastructure.

DINAC is a provider of key data on the state of WASH infrastructure in schools. These data can feed into the information systems of the CNAEHA or the provincial directorates of education, in particular on: *the number of schools with access to drinking water, the rate of coverage of functional latrines per student, the state of functionality of handwashing facilities, the need for new infrastructure or rehabilitation*.

- **Directorate of Life Education, DEVC**

The mission of the DEVC is to promote civic and behavioural education, including the adoption of good hygiene practices in everyday settings, in schools and in the community. It works to anchor the values of citizenship, health and the environment in the behaviour of students and teachers.
Its key missions are: integration of hygiene, sanitation, citizenship and water management modules into school curricula, the development of educational content and awareness-raising tools (posters, guides, monitoring booklets), the training of teachers and school clubs in WASH, coordination with NGOs and partners in the implementation of hygiene awareness campaigns in schools and communities, support for the activities of school hygiene clubs (theatre, cleaning, handwashing campaigns, etc.).

The DEVC produces and manages qualitative and quantitative data on educational behaviours and campaigns: number of schools with an active hygiene club, frequency of awareness-raising activities carried out in schools, level of knowledge and practice of pupils on hygiene behaviours.

- **Current state of information management**

DINAC has begun to structure its information management system functionally. The data are collected in the form of structured Excel files, which are filled in and updated regularly by the provincial branches. These files are then transmitted to the central level at a defined frequency (monthly or quarterly depending on the province), which allows for continuous technical monitoring of the state of school infrastructure at the national level. The data collected covers a range of essential elements related to schools, including: the type of building materials (durable,

semi-durable, precarious), the number of classrooms per school, the number of students disaggregated by gender, the presence or absence of functional toilets, the number of doors in toilet blocks, the availability of handwashing facilities.
While this data is well-organized and actionable for internal needs, DINAC does not yet have an automated IMS or a dynamic consolidation or visualization platform. Files remain stored locally or transmitted by email, without a centralized national database or formal mechanisms for interoperability with WASH IMS (such as those of the CNAEHA or the DHSP).

The DEVC, on the other hand, presents a more rudimentary situation in terms of information management. Data are collected in the form of paper reports or Word files, produced at all levels (schools, inspections, provinces), without uniform structuring or systematic transmission to the central level.
These reports mainly provide information: the activities carried out by hygiene clubs or teachers (theatres, awareness campaigns, cleanings), specific initiatives to promote civic and hygienic behaviour, feedback on the implementation of daily life modules in schools.
However, the lack of a standardized canvas, a digital database, or a centralized platform makes overall analysis very difficult. The information is neither harmonized nor consolidated, and its use remains limited to administrative documentation purposes rather than results-oriented management.

In addition, within the Ministry of National Education and New Citizenship, other directorates already benefit from a robust information management system, known as E-GIS. This system, administered by the Directorate of Statistics and Planning
(DIGEP), constitutes the central core of the information architecture of the education sector. It hosts key modules such as: the school yearbook, listing all schools across the country, the school map, integrating geospatial data on location and educational coverage, school performance reports, filled in on a semi-annual basis.

The E-GIS server is hosted within the ministry, and the data is regularly fed by the provincial statistics divisions, with a good level of structuring, security and accessibility. The system is also interoperable, with the possibility of exchanging data with other sectoral systems.

In this context, two advocacy actions must be undertaken for the integration of the specific indicators produced by DINAC and DEVC within the E-GIS. This integration would make it possible to structure all information related to school infrastructure (DINAC) and hygiene and life education (DEVC), in line with sector standards. Finally, the CNAEHA IMS provides a functional connection with the E-GIS to retrieve WASH data in schools, thus facilitating multisectoral dialogue and joint analysis of the performance of basic services in schools.

8. Project and Programme Monitoring Unit, CSPP

The CSPP is a technical structure within the Ministry of Water Resources and Electricity, responsible for ensuring the strategic steering, technical coordination and monitoring and evaluation of the implementation of sectoral projects and programs, in particular those financed by technical and financial partners (TFPs). It is an essential lever for the operational governance of the water sector in the Democratic Republic of Congo.

Key roles and missions

- Monitoring of the technical, administrative and financial progress of investment projects. o Coordination with Project Management Units (PMUs), donors and implementing structures.
- Regular production of monitoring reports, dashboards, and accountability documents.
- Support for the consolidation of data from technical directorates, provincial boards and partners.

- **Current state of information management**

Currently, the CSPP uses a combination of digital tools to manage information related to ongoing projects and programs.

Klastoo Platform

The CSPP centralises the majority of its files (technical documents, reports, internal memos, official correspondence) and emails on the Klastoo collaborative platform, which allows thematic archiving by project, secure management of access rights and structured sharing between authorised members. However, this solution is still primarily oriented towards document storage and does not replace a real analytical or data visualization system.

Power BI dashboard integrated with an internal website

For the dynamic monitoring of projects, the CSPP has developed an interactive dashboard using Power BI, hosted on an internal website accessible to authorized members of the ministry. This dashboard provides real-time visualization of project progress data (technical milestones, execution rates, disbursements, delay alerts), and represents a significant step towards a culture of accountability based on evidence.

Despite these advances, the current system remains partially compartmentalized, with no direct interconnection with the information systems of the technical departments (SG-MRHE, RPSPE, CNAEHA) or technical and financial partners. In addition, the feeding of dashboards is still mainly based on the manual transmission of Excel files, which can cause update delays, duplicates or consolidation errors.

To strengthen the effectiveness of monitoring and improve responsiveness, we recommend that the CSPP open a dedicated interface for implementing units and partners. This interface would allow the "push" of monitoring data directly to the platform, via a secure web interface for encoding or importing standardized files (Excel, CSV) and the connection of technical departments for the pull of financial data into the analyses and dashboard. Such a development would facilitate the real-time reporting of information, improve the reliability of data, and strengthen the link with existing information systems within the ministry and the sector.

9. The water distribution authority, REGIDESO

The Régie de Distribution d'Eau de la République Démocratique du Congo (REGIDESO SA) is the public company responsible for the production, treatment and distribution of drinking water

in urban and peri-urban areas in the Democratic Republic of Congo (DRC). Under the supervision of the Ministry of Water Resources and Electricity (MRHE), it serves more than 100 urban centers, operating an extensive network of infrastructure including capture, treatment, pumping, storage stations and thousands of kilometers of pipelines. REGIDESO's mission is to ensure a reliable and sustainable supply of drinking water, thus contributing to the well-being of the population and the socio-economic development of the country.

To support its modernization and digitization efforts, REGIDESO has a specialized Information Systems Department (ISD). This directorate is structured into three main divisions:

- *Networks and infrastructure* : responsible for the management and maintenance of IT infrastructure and communication networks
- *Applications and Development* : Responsible for the design, development and maintenance of internal software applications
- *Support and User Assistance* : provides technical support and assistance to users of the company's IT systems

This organization aims to ensure effective management of information technology within REGIDESO, thereby supporting the various digital transformation initiatives.

- **Current state of information management**

REGIDESO has undertaken several digitalization initiatives to improve its operations and the quality of its services. These initiatives are currently in the pilot phase and have not yet been deployed in the provinces concerned by phase 1 of the Water and Sanitation Services Improvement Programme (PASEA).

Computerization of Internal Processes via an ERP

REGIDESO is working on the implementation of an integrated management system (ERP) covering all its internal functions, including subscriber management, accounting, human resources, maintenance, logistics and asset management. This ERP aims to centralize operations and provide real-time management data, improving the traceability of interventions and decision-making. Currently, this platform is in the test phase in some pilot centers.

Water Prepayment Payment Devices

A prepayment initiative has been launched, involving the installation of rechargeable card terminals and mobile payment systems. This system would allow users to credit their consumption in advance, thus reducing unpaid bills and facilitating collection.

This solution is currently being tested in a few urban areas.

Installation of Smart Electronic Meters

REGIDESO is piloting the introduction of smart electronic water meters capable of transmitting the volumes consumed remotely. This initiative aims to automate invoicing, limit fraud and improve the accuracy of measurements. The pilot program includes tests on interoperability with payment tools.

Network Review and Installation of Sensors for the Reduction of Non-Revenue Water A technical review of the network is underway to reduce the rate of non-revenue water (ENF). This initiative includes the installation of pressure and flow sensors, leak analysis and real-time monitoring of losses on the network. The aim is to better understand the causes of losses and to plan targeted repairs.

Development of an Integrated Digital Platform

REGIDESO is developing an integrated digital platform aimed at centralizing interactions with subscribers, customers, suppliers and other stakeholders. This platform will allow for online

referrals, emergency reporting and requesting services, reducing wait times and improving operational efficiency.

Online Bill Payment

REGIDESO is in the trial phase to evaluate online invoice payment solutions. Subscribers will be able to pay their bills via online banking platforms (eBanking), mobile phones, credit cards or branches. This initiative aims to make the payment process easier for customers and improve debt collection.

Application of Data Science and Artificial Intelligence

REGIDESO plans to apply data science, artificial intelligence and machine learning to analyze operational data. The goal is to learn from the data collected, learn from it, and improve operations and customer experience.

In the regional directorates of Mbuji-Mayi, Kananga, Bandundu, as well as at the operational center in Tshikapa, operational information management is still largely based on a hybrid system, combining paper media and rudimentary digital tools. The data is collected regularly on paper cards, then manually encoded into Excel files, before being transmitted via the RegiDocs application, which serves as a centralized digital space for saving and archiving reports.

The types of data collected cover most field operations, including: ○ *Production volumes and plant performance* ○ *Monitoring of the quality of the water produced* ○ *The operation of the generators and the consumption of fuel or other inputs* ○ *The situation of breakdowns and the maintenance interventions carried out* ○ *The technical situation of the distribution network*

The frequency of submission varies according to the indicators: critical operating data is transmitted daily, while other parameters (such as maintenance or consumption) are reported on a weekly or monthly basis. This mechanism allows for regular feedback, but is still dependent on a manual process that is not interoperable with other systems, limiting real-time analysis and the systematic use of data for strategic management purposes.

A major gap in REGIDESO's information management system is the lack of a formal governance framework for sharing data with other actors in the water sector, both at the national and provincial levels. To date, there is no institutionalized mechanism defining the types of information to be shared, their frequency, or the modalities of access to the data generated by REGIDESO. This gap severely limits sectoral coordination, integrated planning, as well as joint monitoring of the performance of the public water service.

In particular, data exchanges between REGIDESO and central bodies such as the CNAEHA, the General Secretariat of the MRHE (SG-MRHE), and the Regulatory Authority for Public Water Services (ARSPE) are ad hoc and not governed by standardised protocols or conventions. No mapping of information flows has been established, nor an agreed frequency of transmission, which prevents a consolidated reading of the sector's performance and limits the implementation of the principles of transparency and accountability.

To fill this gap, it is essential to develop a national sectoral data governance framework, under the aegis of ARSPE, which will specify the roles, responsibilities and sharing obligations for each public or private operator. This framework will also need to be integrated into the ARSPE's Information Management System (IMS), to enable the automated collection, cross-validation and consolidated analysis of operational, performance and quality of service data provided by REGIDESO and other delegates

10. Public Water Sector Regulatory Authority, ARSPE

The Public Water Services Regulatory Authority (ARSPE) is an institution provided for by Law No. 15/026 of December 31, 2015 on water, and officially created by Decree No. 22/04 of March 1, 2022. It is currently being implemented in the DRC.

ARSPE is a key player in the governance of the water sector, with the main mission of guaranteeing the transparency, performance, and quality of the public water service, while ensuring that the rights of users and the contractual obligations of operators are respected. According to the legal texts, its missions include: o *Ensuring compliance with concession contracts, authorisations and declarations from operators*

o *Monitor the application of norms and standards by public water service operators*

o *Establish specifications and normative documents for the allocation of concessions* o *Ensure conciliation in the event of disputes between operators and between operators and users* o *Supervise pricing: determine the rules for setting, analyse tariff proposals,* o *Monitor the functionality, financial and technical viability of water services*

To carry out its mission, ARSPE will have to equip itself with an IMS system with the following critical data:

Type	**Description**	**Indicators**	**Collection Frequency**
Demographics	Population by administrative level, population served, number of connections, coverage area	% of water coverage	Monthly
Concession Contracts	Contract type, coverage area, operator	Number of contracts Assets by province % of contracts that comply with standards	Semi-annual
Compliance with service and quality standards	Directory of standards, water quality monitoring,	% Standard-compliant structure Rate of service quality complaints	Quarterly
Type	**Description**	**Indicators**	**Collection Frequency**
		Number of operators in violation of standards on water quality	
Pricing	Cost structures, subsidies, tariffs charged	Number of tariff proposals approved or rejected Average time to scan Pricing Number of cases of	Semi-annual

		Violation of approved tariffs	
Financial Management	Financial statements, recovery ratio, level of debt, operators	Recovery rate Medium Cash flow ratio Number of viable operators	Quarterly
Network and distribution management	Failure frequency, level of preventive maintenance, ENF	Average number of days service interruption Technical availability rate % of NFTs	Monthly
User satisfaction	Date of complaints, description of complaints, date of escalation of complaint, classification of complaint, date of resolution of complaint, results of satisfaction surveys	% of complaints closed within the deadline Average satisfaction survey score	Monthly/Yearly (Satisfaction Survey)
Litigation and litigation	Disputes, Dispute Description, Operators Involved, Dispute Resolution Level, Case Close Date, Dispute Resolution Summary	Number of cases resolved amicably Average time to resolution	Quarterly
Regulatory framework	Classification of the standard, reference of the standard, abstract, pdf document of the standard, date of update or publication of the standard, alignment note to regional standards	Number of standards in Vigor Average time Refresh Number of standards in need of alignment with regional standards	Annual
Funding and ongoing projects	Amounts mobilised, status of projects, donors involved	Mapping of Priority Areas for Funding [disaggregated by administrative level] Gap in financing [disaggregated by level Administrative] % of projects at risk in implementation	Quarterly

Table 17: Critical Data for EPSRA

This indicative data framework will allow ARSPE to set up a structured monitoring system and feed into its future information management system (IMS).

The current situation, marked by the gradual implementation of the ARSPE, represents a major strategic opportunity to structure from the outset an Information Management System (IMS) adapted to its regulatory mandate. By integrating the digital dimension into the design of its tools and procedures, ARSPE can establish a robust, interoperable data architecture aligned with the requirements of transparency, accountability and benchmarking of performance. This founding moment thus offers the possibility of defining a standardised framework for the collection, validation, archiving and sharing of data with all operators and sectoral institutions (CNAEHA, SG-MRHE, RP-SPE), consolidating its role as a national pivot for evidence-based regulation

2. Inventory of existing data

The data inventory carried out as part of the diagnosis of the information management systems (IMS) of the water and sanitation sector made it possible to compare the data currently available with those needed for the monitoring of the indicators of the PASEA results framework.

The main findings from this exercise are as follows:

1. Existence of IMS within Interest Structures

At the national level, only *three of the eleven organizations visited have structured IMS integrated into their business processes* : the DRE, the CSPP and the REGIDESO. These systems allow for effective internal management of critical data, but their integration into the sector's coordination and regulatory mechanisms remains limited. Information sharing between institutions is almost non-existent.

At the provincial level, *three of the fourteen organizations visited have an IMS* (SPD, ESPP and TSEP). However, these LMS only partially cover the data that is critical to managing the WASH sector, and in the case of EPSRs, they are not yet fully integrated into day-to-day business processes. *Four other structures use Excel files as their main data management tools* : the regional directorate of REGIDESO, the provincial directorate of ONHR (only in Kasai Oriental), the Hygiene Division, and the CPEDD (for data relating to the FDAL roadmap). *The remaining seven organizations, on the other hand, keep the data in the form of paper reports or Word files*. With the exception of the procurement unit, none of them has an adequate archiving system. The exploitation of historical data is therefore extremely difficult, if not impossible.

National level	
CNAEHA	No structure-level IMS
SAR	An Excel-based IMS that covers data on the FDAL roadmap
DESP	NA
DHSP	No structure-level IMS. Critical elements on water and sanitation not included in DHIS2 used as SGI in the health sector in the DRC
DEVC	No structure-level IMS. Difficult to use Word or paper report archiving system
DINAC	No IMS at the level. Internal efforts to structure data on Excel databases in progress
DRE	Several structured and integrated IMSs in the Great Lakes region as well as in the Nile Basin countries under the management of the

	Hydroinformatics Centre: River level trend data, weather forecasts and a number of models on the river basins and sub-basins
ONHR	A structured Excel-based IMS that covers water infrastructure data. Not integrated database on all provinces and many data still in Word reports
SG RHE	Partial provincial databases on mWater with water infrastructure mapping data
CSPP	A structured IMS integrated into business processes with a focus on document management and a PowerBI dashboard for monitoring projects and programs. A good digital culture
REGIDESO	Several internal and integrated IMS are in the pilot phase but not yet deployed. Organization in the process of digital transformation
Provincial level	
Division Provincial de la sante	DHIS2 as a healthcare SGI. Lack of data and some key indicators on the water and sanitation sector
Provincial Water Utilities Board	mWater as an SGI but not integrated into all the business processes of daily management. mWater not yet implemented in the provinces of Kasai and Kasai-Central.
ONHR Provincial Directorate	No structured IMS. Documents in Word format except for the province of Kasaï-Oriental which has databases in Excel format
CPEDD	No structured IMS. Data in the form of Word reports without an archiving system except for monitoring the implementation of the FDAL roadmap
Hygiene Division	An Excel-based IMS that covers data on the FDAL roadmap
DEVC Branch Coordinator	No IMS
Division EPST	E-GIS as SGI at the provincial level with the school map and the yearbook. Need for details and disaggregation of data on the water and sanitation sector
Regional Directorate, REGIDESO	SGI based on Excel and REGIDOCs files. Pilot national IMSs not yet deployed in Kasai, Kasai-Central, Kasai-Oriental and Kwilu provinces
Governor	NA
Water Resources and Energy Division	No structured IMS. Data management based on paper reports
CPAEHA (CENAEHA Province)	No IMS
DINAC	No IMS at the provincial level. Paper files without any clear method of archiving and sharing data at the national level. Only the activity reports submitted each quarter
Public Procurement and Management Unit	SGI based on paper documents managed in a structured way with a robust archiving system
Provincial Minister of Budget and Finance	NA

Table 18:Inventory of existing IMS

2. Status of data in relation to needs for the indicators of the ESAP Results Framework

Existing IMS do not have the data to inform the indicators in the ASEAL Results Framework.

Component/Results/Indicators of the PASEA	**Comment on the availability of the data needed for the calculation**
People with access to at least basic drinking water services (Number (thousands))	Partial data available within the RPSPEs and hosted on mWater: analysis needs by coupling the location of water infrastructures with population data

People with access to basic sanitation services (Number (thousands))	Partial data available only in rural areas in the Excel databases for monitoring the ODF roadmap. Data
	hosted by the DAS, DHSP, Hygiene Division or the CPEDD
Strengthening the capacity of the private sector to provide safe drinking water and sanitation services	Partial data available on two of the four result indicators see the operators' performance report on mWater
Improving access to drinking water supply and drinking water distribution capacities	Data not available on the nine indicators in existing IMS
Improved access to and capacity to deliver sanitation services	Data not available on the nine indicators in existing IMS
Persons who participated in the Young Professional Programme organised within the Provincial Implementation Unit of the Project (Number)	Mechanism in place for the availability of data necessary for the management of UPEP activities
Complaints Addressed and Resolved in a Timely and Satisfactory Manner (Percentage)	Partial data available on the conduct of the programme according to the implementation mechanisms. Data on complaints in water and sanitation services not available in existing SGIs in the sector.

Table 19: Status of Data Availability for ASEA Indicators in Existing IMS

3. Preliminary roadmap

This roadmap is an extension of the study of existing information management systems (IMS) within the water and sanitation sector in the Democratic Republic of Congo. This analysis highlighted a proliferation of fragmented solutions, often implemented autonomously by institutions such as REGIDESO, DAS, SG-MRHE, RPSPEs or CNAEHA, without a unified governance framework or interoperability. In this context, the establishment of an integrated, unified and inter-institutional IMS is a priority to strengthen the coordination, planning, monitoring and regulation of the sector at all levels (national, provincial, local).

To ensure its effectiveness, this IMS must be designed from the outset around a clear governance framework, involving the key players in the sector: ARSPE, CNAEHA, DAS and RPSPE. It must also be based on the standardization of existing tools, in particular the collection frameworks currently used in a heterogeneous way by technical departments (e.g. the same demographic, infrastructure or quality of service data being collected in different formats). The production of harmonised data collection and submission tools is therefore essential to ensure the integration of information in the same digital environment. Finally, this roadmap draws lessons from regional and international good practices, in order to adapt proven solutions to the institutional, technical and financial reality of the DRC.

4. Proposed IMS Governance Framework for Water and sanitation in the DRC

Actor	Role in IMS	Proposed structural integration
ARSPE (French Broadcasting Authority) Regulation of Public Water Services)	Regulator: requires large public and private operators to share critical data in pre-established formats and deadlines	Integration of SGI reporting requirements into contracts and regulatory mechanisms
CNAEHA (National Water, Hygiene and Sanitation Action Committee)	Coordination and sectoral steering: integrates data from ministerial directorates and institutions (DAS, ONHR, REGIDESO, etc.)	IMS becomes the official source for multi-sectoral analysis and reporting
DAS (Sanitation Directorate)	Public technical operator: integrates the IMS into its tools for planning, monitoring and managing the FDAL roadmap	Adoption of IMS as a standard business tool for its four technical functions

RPSPE (Provincial Regimes) of Public Water Services) – SG-MRHE	Adopts the IMS as an operational tool for service management, contracting with operators, supervision and reporting	The IMS becomes the official platform for the management of contracted operators and performance indicators

Table 20: Proposed governance framework for IMS in the water and sanitation sector in the DRC

The governance framework is designed to support the implementation of IMS during the PASEA program while ensuring its institutional sustainability. During the implementation phase of the programme, it benefits from the technical and logistical support of the Project Implementation Unit - Water (CEP-O), in particular through the training of users, the coverage of accommodation costs and the regular evaluation of the deployment of the IMS. This framework is designed to achieve full autonomy at the end of the PASEA, thanks to the establishment of a pool of technical experts in information systems management and maintenance, data analysis and functional administration, trained and integrated within the directorates and member institutions (ARSPE, CNAEHA, DAS, RPSPE). This approach aims to ensure continuity of technical support, national ownership and sustainability of the system beyond external support.

5. Standardization of data and production of an SGI-EHA collection guide

Data standardization is essential to ensure consistency, integration, and efficient analysis of information in the EHA industry's IMS. Currently, directorates and institutions collect similar data in various formats, making it difficult to use them together. The adoption of harmonised frameworks and frameworks will improve the quality, comparability and consolidation of data at the national level. The table below outlines some categories of critical data to standardize in conjunction with key industry players

Domain	Data type to be standardized	Tools/Standards to be integrated
Demographics & territorial	Population, administrative level, coverage of services	Standards: INSTAT, PNEHA, official administrative code
Infrastructures EHA	Boreholes, latrines, networks, stations, waste management	Standardized sheets (location, type, condition, manager) with photo/GPS guide, technical characteristics, and.
Water quality	Results of microbiological and physicochemical analyses	National standards related to DHSP, WHO guidelines
Actors & Operator	Identification, intervention area, status, management model, classification of interventions	Unique identification code, harmonised register, etc.
Common indicators	Access to water, improved latrines, continuity of service, non-revenue water, recovery rate, coverage rate	Harmonization of calculation formulas, validation with ARSPE/CNAEHA
Financing & projects	Budget mobilised, progress status, donors, area covered	Project scorecards aligned with the CSPP framework

Table 21: Critical data areas to be standardised between managers and actors in the WASH sector

The standardization of critical data will produce a national guide for the collection and submission of IMS-EHA data including the frequency of data transmission, and templates for importing existing databases.

6. Study of good practices at national, regional and international levels

The study of good practices is based on three relevant cases to guide the design of the SGIEHA in the DRC.

- **mWater**

It is already used within the RPSPE, constitutes a useful national reference, demonstrating the technical feasibility of digital and geolocalized collection in the Congolese context, despite the lack of consolidated governance.

- **NWASCO and RUWASA SDMS**

The experiences of Zambia (NWASCO) and Tanzania (RUWASA SDMS) are particularly inspiring because they come from institutional and logistical contexts similar to that of the DRC: they show how a clear regulatory framework and decentralized management can ensure the performance of a multi-source IMS.

The implementation of the integrated, unified and inter-institutional IMS-WASH will continue in three key phases. First, the validation of the proposed governance framework by stakeholders (CNAEHA, ARSPE, DAS, RPSPE) in order to ensure buy-in and clarity of roles. Secondly, the joint work of this governance framework for the standardisation of collection tools and data formats, an essential condition for interoperability. Finally, the conduct of targeted studies on national, regional and international good practices to enrich the final design of the system.

CHAPTER 2 : STUDY OF GOOD PRACTICES NATIONAL AND INTERNATIONAL

In this chapter, we presented the best practices resulting from the implementation, maintenance and evolution of EMS-EHA at the local, regional and international levels.
We also analyzed the recommendations of the national ministry in charge of digital technology for alignment with the country's vision in the digitalization of services and information management,

1. National Good Practice: mWater in DRC

mWater is used as an SGI in the RPSPEs in the provinces of Kasai-Oriental, KasaiCentral, Kasai, Kwilu, North Kivu, South Kivu, Lomami and Sud-Ubangi. It was introduced in the DRC as a tool for geo-referenced collection of water points for the first time in 2015 as part of the sanitation schools and villages projects. The poor connectivity in the country for smooth data transmission and its non-integration with DHIS2 will be the basis of its decline around 2018.

mWater will be reintroduced in the country with the RISE (Institutional Strengthening of Water Services) project, implemented by GIZ in collaboration with the Ministry of Water Resources and Electricity (MRHE) in 2020, aimed to strengthen the planning and regulation of public water services in several pilot provinces, including Kasai-Oriental, Kwilu and Lomami.
As part of this, mWater has been introduced as a strategic tool to:

- Map water points and associated infrastructure (boreholes, wells, reservoirs, standpipes, etc.)
- Evaluate their functionality and level of service.
- Create a dynamic database to inform decision-making at the provincial and national level

Its technical device includes: the mWater surveyor mobile application for data collection and the web interface with mWater studio for the creation of forms as well as mWater Dashboard for analyses. At the current stage, the autonomous databases are being deployed in the provinces and not centralized in a unified database at the national level.
The technical management support team mentioned the following lessons learned and good practices for the implementation of IMS in the WASH sector in the DRC:

1. Integrate the sustainability model from the design phase

The mWater approach in the RISE project has integrated sustainability at two strategic levels:

- **Reduced operational costs**, including no system hosting costs, which ensures continuity of access to data after project closure.
- **Targeted training of technical teams**, in particular focal points within the RPSPE and the MRHE, to ensure the autonomous management of the tool.

In a context such as that of the DRC, where many public structures are struggling to mobilize operating budgets, the adoption of tools without recurring costs (hosting, licenses) is essential. Many systems developed in the past have been abandoned, data lost as a result of the expiration of hosted services. The example of the **DHIS2 instance dedicated to healthy schools and villages** is an illustration of this.

At the same time, the establishment of a **technical pool** formed within the user structures has made it possible to ensure the maintenance of activities and the continuity of the tool. This transfer of skills reduces the need for external support for basic maintenance, reducing costs and the risk of downtime. Conversely, initiatives such as **the STUDY system of the CNAEHA** have failed due to a lack of trained staff to manage them after the end of funding.

2. Include the on and offline working mode when collecting data

The first attempt to introduce mWater in the DRC, in 2015, was a partial failure due to **the lack of a functional offline mode**. Poor internet coverage in rural areas has led to the loss of a lot of data and discouraged field teams.

The evolution of mWater, with a **powerful offline collection mode**, made it possible to collect the data in the field and then to synchronize it later, once the connection was available. This feature has become an **essential prerequisite** for any IMS to be implemented in contexts with limited connectivity. It will have to be natively integrated into future modules, taking into account the **specific architecture of each system**.

3. Co-create the collection tools with stakeholders

One of the successes of the RISE project lies in the **co-construction of collection tools** with end users, in particular the Provincial Water Utilities Boards (RPSPE). This participatory approach has made it possible to develop forms adapted to the realities on the ground and to the operational needs of public structures.

This **functional alignment** reinforced the appropriation of the tools, preventing them from being perceived as products exclusively related to the project. As a result, their use was able to continue even after the funding ended. This strategy reduces the risk of systems being abandoned in silos or without direct user involvement.

2. National Information System for Water Supply and Sanitation (NISWSS)

The National Water Supply and Sanitation Council (NWASCO) was established in 2000 as a result of Zambia's water sector reform that began in the 1990s. Its creation stems from the Water Supply and Sanitation Act No. 28/1997, with the aim of regulating drinking water and sanitation service providers in urban and peri-urban areas

Its main mission is to regulate water supply and sanitation services to ensure efficient, equitable and sustainable delivery, while protecting consumer rights. The MWASCO has defined a mandatory nimumim service level for all commercial and private providers based on water quality, number of hours of service, billing, customer service, service disruption planning, pressure on the water network, unjustified disconnections, and wastewater treatment.
In 2015, NWASCO launched the **National Information System for Water Supply and Sanitation** (NISWSS). Its technical component includes a web application with three different components: *a database of all information collected from commercial and private service providers in the sector, an integrated community monitoring system in the form of local committees authorized to report service failures and an open web portal with GIS functionalities allowing cartographic access to data by district, by operator, or by indicator*.

The NISWSS is fed monthly and annually by data submitted by commercial and private providers and then validated by NWASCO inspectors. The NISWSS is also fed by data from NWASCO's service operations such as the allocation of coverage areas, the validation of rates, the reception and processing of complaints as well as inspection and control missions.

Commercial and private service providers submit data via an interface on twenty indicators that allow benchmarking.

Indicator	**Description**
Non-Revenue Water	Percentage of produced water that is not billed (leaks, thefts, errors)
Water quality	Percentage of samples that meet quality standards
Metering Ratio	Proportion of connections with working meters
Water Utility Coverage	Percentage of the population with access to safe drinking water
Sanitation coverage	Percentage of population with adequate sanitation
Water supply hours	Average number of hours per day of water supply
Staff Efficiency	Number of employees per 1000 connections
Efficiency of collection	Percentage of billed revenue actually collected
Percentage of billed revenue actually collected	Percentage of billed revenue actually collected
Total cost recovery rate	Total cost coverage level (O&M + investments)
Connection growth rate	Annual rate of increase in connections
Response time to customer complaints	Average time to handle customer complaints
Customer satisfaction rate	Level of consumer satisfaction with services
Infrastructure renewal rate	Proportion of infrastructure replaced or renovated.
Wastewater treatment rates	Percentage of wastewater treated according to standards
Energy efficiency	Energy consumption per m³ of water produced or treated
Physical losses	Physical losses
Business Losses	Losses due to billing errors or theft
Average time to repair failures	Mean time to repair after incident
Mean time to repair after incident	Measure of compliance with regulatory requirements

Table 22: Indicators collected in the NISWSS

The twenty indicators are grouped into nine for a classification of the annual performance of commercial and private providers: NFE (25%), water quality (95%), metering rate (100%), water service coverage (95%), sanitation service coverage (85%), water supply hours (18%), staff efficiency (0.4), percentage of billed revenue actually collected (90%) and total cost recovery rate (100%).

After ten years of implementing the NIWSS, the NWASCO recommends:

1. Strong legal framework

Regulation by NWASCO is backed by a clear law, with sanctioning prerogatives. The data and frequency of submission is part of the legal framework. This strengthens the accountability of commercial and private providers in the submission of data.

IMS-EHAs should include the required data from providers in the contractual framework with commercial providers with clear penalties for non-compliance with data submission frequencies and timing.

2. Industry standardization

The use of indicators common to all providers, supported by technical manuals and guides, facilitates data interoperability.

3. Integrated Community Participation

The local committees authorised to report service failures provide citizen monitoring in addition to the technical analysis.

4. Continuing education for commercial and private service providers

Training ensures the ability of providers to report data in a complete and consistent manner. The team service providers ensure the continuity of reporting.

3. RUWASA Service Delivery Management System

The RUWASA SDMS was introduced in Tanzania in response to persistent challenges in monitoring and planning water, sanitation and hygiene services, including: poor coverage of reliable data, fragmentation of information systems, and lack of standardization across ministries and administrative levels.

The first version was developed in **2008** as part of the reform of the **Water Sector Development Programme (WSDP),** with the support of the World Bank, GIZ and other technical and financial partners.

Since then, the RUWASA SDMS has gone through **several iterations**, culminating in the current interoperable version launched around **2017–2018**, capable of **synchronizing district data with national systems** via the internet and mobile tools

The RUWASA SDMS aims to:

- **Centralize data collection** on water, sanitation and hygiene infrastructure,
- **Standardize the monitoring of sector KPIs** (access, functionality, quality),
- **Facilitate planning, budgeting and accountability** at the national, regional and local levels,
- **Ensure regular reporting towards the SDGs** and other international commitments (e.g. JMP WHO/UNICEF).

Its technical system includes: a central database hosted at the Ministry of Water and interconnected with regional databases, standardized data collection forms on water points, sanitation facilities, quality of service, local governance and the GIS viewer of interactive maps of WASH infrastructures with functional statuses by village or district.

The data is populated by the district engineers via a mobile application that has both online and offline collection capabilities. They are analyzed and visualized via the web portal for automatic production of dashboards and reports. The central database is interoperable with other systems such as DHIS2 for health data, LGRCIS for local finance, and MIS for education

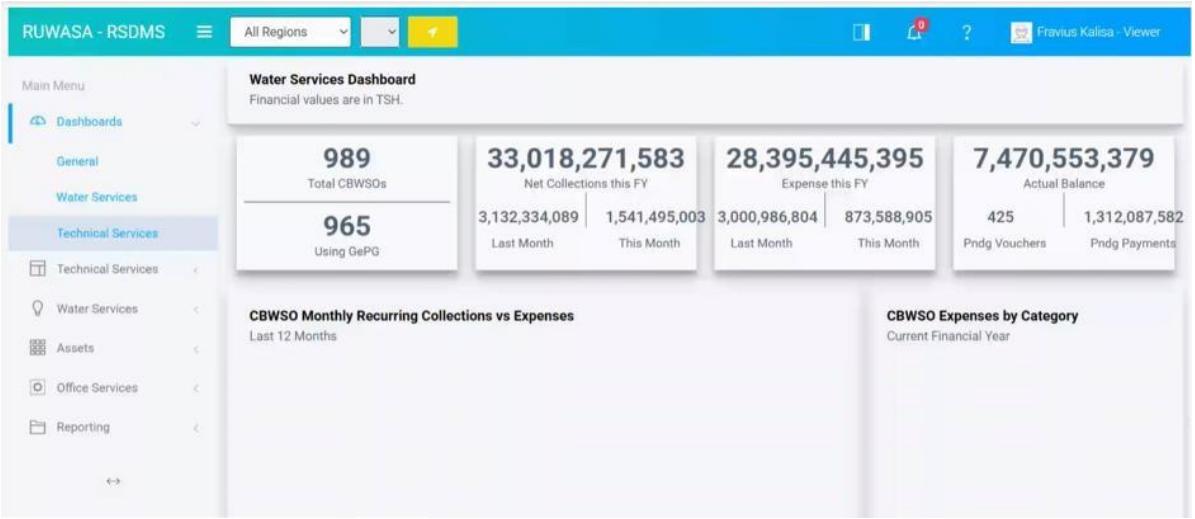

Figure 2: Overview of the RUWASA SDMS

At the current stage, the RUWASA SDMS contains around 90 thousand water points and 10 thousand schools updated between six and twelve months and an alignment of the indicators with the SDGs.

In twenty-five years of management and evolution, the RUWASA SDMS management team has documented the following recommended best practices in the design and implementation of EHA-IMS:

1. Design the system with evolution and interoperability

The RUWASA SDMS has undergone several changes but still with data integrations that allow to preserve the history but also interconnection with the other SGIs in the ecosystem.

The team recommends third-party architecture for systems under design. This makes it possible to design the systems in the form of microservices that communicate with each other via APIs. It therefore becomes easy to create new services and interconnect them with existing ones through APIs.

2. Offline collection mode

The offline collection mode ensures the use of the SGI even in remote areas without internet coverage.

3. Enhance transparency

The public portal allows **for accountability to users**, with accessible dashboards. Some of the custom-sorted data needs to be publicly available to engage the community with IMS.

4. Institutional management

Organize technical assistance sessions from the beginning of the design or implementation to ensure institutional approval. When an institution or project supports state organizations in the design and implementation of IMS, the latter will have to ensure that the services of the state organizations involved take the lead from day one: validation of design choices, review of sprints for deliverables as well as the organization of showcases for the follow-up of feedback and adjustments.

4. Recommendations of the Ministry in charge of digital in the design of IMS in the DRC

The Democratic Republic of Congo (DRC) has embarked on an ambitious digital transformation, guided by the National Digital Plan – Horizon 2025 (PNN). This strategic plan is based on four fundamental pillars:

- **Infrastructure:** development of high-speed networks and secure data centers.
- **Content** : promotion of the local digital industry and data valorization.
- **Application uses** : digitalization of public services and strengthening of digital culture.
- **Governance and Regulation** : establishment of an appropriate legal framework and guarantee of cybersecurity

In the context of the implementation of the PNN, the **Ministry of Posts, Telecommunications and Digital plays** a central role. Its missions include:

- *Strengthening digital governance* : Optimizing postal and telecommunications management to stimulate sustainable growth and improve the impact of digital technology on the national economy.
- *Promoting digital transformation:* Encouraging digitalization and modernizing postal infrastructure to support economic growth and improve public services.
- *Digital literacy development:* Foster the adoption of digital skills to strengthen digital inclusion and accelerate technology integration within society.

- *Creation of a competitive telecommunications and ICT market: Establish* an open and competitive market for telecom networks and services, ensuring that users have diversified choices and accessible innovations.

The digitalization of public services and the development of IMS are part of pillar three of the national digital plan and must be aligned with the recommendations of the ministry.

1. Interoperability of IMS with existing IMS

The solutions developed must be compatible with existing infrastructures and promote the exchange of data between the different government platforms.

To meet this requirement, third-party architecture is appropriate. This makes it easier to establish integration channels with existing IMS via APIs.

2. Local data hosting

Government digital solutions must be hosted in national data centers to ensure digital sovereignty and facilitate data management in accordance with local regulations.

To meet this requirement, we offer the choice of hosting the LMS-EHA at the national level with replication servers for data backup in the cloud. The choice of infrastructure will be made and motivated by the firm in charge of development.

3. Alignment of interfaces with the country's graphic charter

Digital solutions developed for government departments must comply with the national graphic charter. This alignment makes cross-platform use more fluid and improves the user experience. It is accessible on the DRC government graphic charter.

To meet this requirement, the firm in charge of development will have to use the graphic charter when designing the interfaces of the SGI-EHA.

In view of the good practices identified at the national (mWater), international (NIWSS and RUWASA SDMS) levels as well as the recommendations of the Ministry of Digital Affairs, the SGI-EHA in the DRC must:

Best practices	**How to integrate into the design of the IMS-EHA in the DRC**
Integration of the sustainability model into the design and institutional commitment	• Operationalize the governance framework of the IMS-EHA proposed in Table 20 of the validation of the technical choice of the option of the IMS-EHA • Direct the costs of hosting and operational costs of the IMS to organizations with the power to mobilize resources: propose that the governance framework validate the integration of costs in the post-program phase at the level of ARSPE, RPSPEs and the CNAEHA • Train the focal points of the organizations that are part of the governance framework to ensure maintenance from the test phase with the technical support of the CEP-O
Interoperability of the IMS-EHA with existing and future IMS, modularity for	• Adoption of third-party architecture for the design of GIEAHS. The key functions are organized in the form of

Integrations of future options and features	microservices with an interconnection core consisting of APIs. APIs interact with microservices to perform business tasks. • The data that is essential for the indicators is included in a master dataset extracted by key functions • The kernel manages user and other IMS access for interconnection • Develop any future needs as a micro-service and then an API of its interaction with the kernel
Offline operation capability	• Program critical modules useful for users at the territorial and provincial level for offline functionality. Critical modules are offered by the firm and validated by the IMS-EHA governance framework
Robust legal framework to ensure data collection over time	• Include the use of processing modules and return data in the contractual and regulatory provisions: between the RPSPEs and the service providers; between ARSPE and commercial service providers at the national level and between DAS and service providers in the sanitation sector
Standardization of the data collected and method of calculation of common indicators	• Validate through the governance framework the standard frameworks for collecting common data between the different organizations, indicators and their calculation methods: this work will produce the guide for the collection and submission of data
Compliance with the graphic charter in the design of user and management interfaces	• Adopt graphics load when designing interfaces
Hosting at the national level	• Adopting a national hosting model with replication and backup servers in the cloud: choices to be made during the final modeling phase

Table 23: Approach to integrating good practices into the design and implementation of the IMS-WASH in the DRC

CHAPTER 3: PROPOSED INFORMATION IN THE WATER AND SANITATION SECTOR IN THE DRC

In this chapter, we discuss the design assumptions and technical specifications for the modeling of the IMS-EHA in the DRC.

The complete modelling as mentioned in the methodological approach in Figure 1 will be done after the validation of the choice of model proposed in the design hypotheses and the technical specifications.

1. Design Assumption

The design proposal for one or more IMS is informed by the diagnosis carried out in Chapter 1 and the good practices identified in Chapter 2 and summarized in Table 23.

The two options considered are evaluated by comparing critical data, internal validation processes for the execution of the business that generates the critical data, the data processing process and best practices as well as a clear analysis framework

Evaluation Criteria	Key questions	Unified IMS	Separate SGIs
Interoperability and integration	Do the water and sanitation business functions share common flows and indicators?	Promotes data and reporting integration	Risk of duplication and effort of Dubbing
Governance and Responsibilities	Are the institutions of governance common or distinct?	Complexity if the supervisory bodies are different	Clarifies The Roles specific institutional
Implementation costs	Does the budget allow for the financing of two SGIs in a sustainable way?	Optimizes upfront and upfront costs maintenance	Double the cost in infrastructure, training, support
Technical and human capacities	Do the structures have sufficient resources to manage one or two IMS?	Pools capacity, easier to manage	Requires separate teams, training parallel
Flexibility & Scalability	Do future needs require Developments sector-specific measures?	Less flexible in the event of sectoral divergence	Adaptable to specific industry innovations
Resilience & sustainability	What happens in the event of a technical or financial malfunction?	Risk Unique of Global failure	Industry-specific resiliency in the event of an IMS failure

Table 24: IMS Option Choice Framework

Based on this framework and the additional considerations outlined in paragraph 2, we recommend a unified IMS.

The following additional elements support the recommendation:

- The two IMS share several critical data: to avoid duplicity, they must be in a single system. This also guarantees the integrity of this data: administrative and demographic, management of service providers and operators, etc.
- A governance framework including both sectors is put in place through the CNAEHA: this guarantees the same validation framework and constitutes a common supervision in the management of information
- Flexibility and scalability are guaranteed even in the case of a single system by the architecture in the form of microservices
- Resilience and sustainability are also regulated by microservices. There is the ability to isolate non-functional micro-services and ensure the continuity of others in case of failures on a part. Server failures are resolved by the replication and backup servers.

2. High-level overview of the proposed unified IMS-EHA

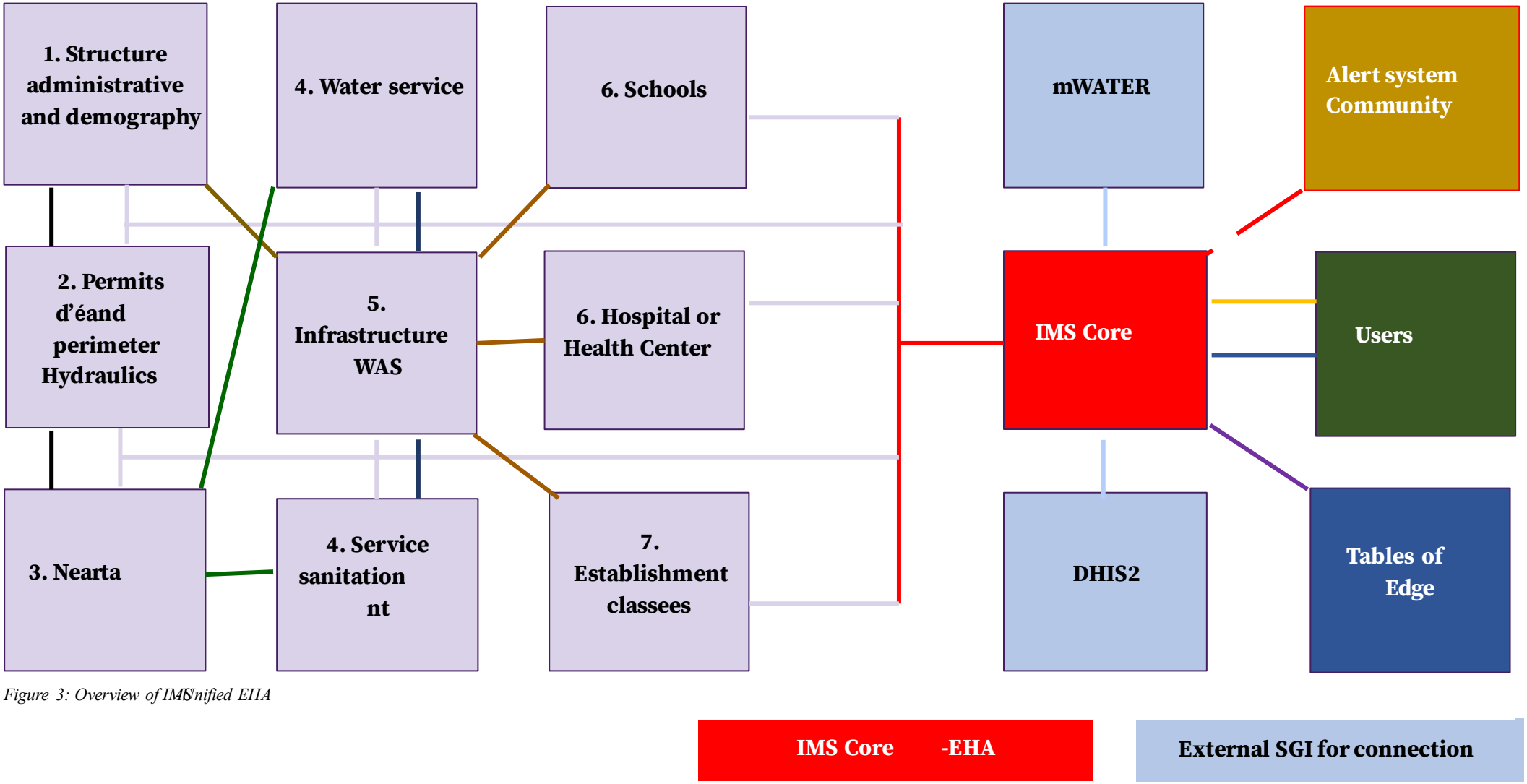

Figure 3: Overview of IMS Unified EHA

	Microservices Management via APIs		Connection and reception of data via external SGIs: mWater pushes water infrastructure into the micro-service, DHIS2 data for inclusion in the calculation of indicators
	Sending and receiving data to microservices		Manage access to microservices to users. Users represent the actors of interest organisations
	Management of the application for study permits and allocation of hydraulic perimeters, delimitation of the area covered by the studies and hydraulic perimeters		Managing user access to dashboards: Each category of users has its own version of the dashboard
	Location of water and sanitation infrastructure (networked or autonomous): administrative location and in a school, classified establishment or health centre		Data transfer from micro-services to the dashboard: coordinate via APIs to the core of the EMS-EHA: includes the operations of calculating indicators
	Assignment of infrastructure to the category: water or sanitation		Community water quality alerts, leak monitoring, sewage infrastructure overflow monitoring, and service quality complaints (time of service availability, delay without pickup and dumping, etc.)
	Provision of water and sanitation services: including the contractual mode of supply, performance monitoring, pricing		

Table 25: Overall view of the flow between microservices, kernel, and connections to external LMS

3. Roadmap

At this stage of the study, the validation of the technical position is necessary to accompany the complete modeling of the IMS-EHA.

I. APPENDICES

- Terms of Reference for the Mission
- Maintenance Guides for SGI Diagnosis
- IMS Modelling Files

Terms of Reference for the recruitment of an (Individual) Consultant in charge of developing Terms of Reference to acquire one or more Management Information System(s) (MIS) on Water, Hygiene and Sanitation (WASH) aspects for the four targeted provinces of the PASEA Project: Kwilu, Kasai, Kasai Central and Kasai Oriental

1. BACKGROUND

1.1. INTRODUCTION

The Government of the Democratic Republic of Congo has received support from the World Bank Group's International Development Association (IDA) to implement the Access to Water and Sanitation Services in the DRC (PASEA).
The development objectives of this program are:

- Increasing access to basic drinking water supply and sanitation services in some provinces of the DRC and;
- Strengthening the capacity of the public and private sectors to provide safe drinking water and sanitation services

This program will be implemented according to the multi-phase programmatic approach, the first phase of which concerns the peri-urban and rural areas of the provinces of Kwilu, Kasai, Kasai Central and Kasai Oriental. Phase 2, whose approval will depend on progress made under Phase 1, aims to expand the intervention to other ETDs among 5 other provinces (Ituri, Kongo Central, Lomami, North Kivu and South Kivu). Phases 3 and 4 will focus on sustaining service delivery, encouraging improved performance by provinces and service providers in all nine provinces.

During the programme period, it is envisaged that water and sanitation infrastructure will be built in the nine provinces and that capacity to manage and monitor services will be strengthened with a view to providing 12 million additional people with access to basic drinking water services and 8 million additional people to basic sanitation services.

The AMAP has four components below.

1. **Improving Access and Capacity to Provide Drinking Water Services**
    1.1. Improving Access to Drinking Water in Rural and Peri-Urban Areas;
    1.2. Improving the Performance of Private and Non-Profit Operators water;
    1.3. Strengthening Public Institutions and Capacities for Drinking Water Services.
2. **Improving Access and Capacity for Sanitation Service Delivery**
    2.1. Improving Access to Sanitation and Hygiene in Rural and Peri-Urban Areas;
    2.2. Improving Access to Water, Sanitation and Hygiene (WASH) in Institutions;
    2.3. Private Sector Development for Sanitation and Hygiene;
    2.4. Strengthening Public Institutions and Capacities for Sanitation Services.
3. **Project Management, Learning and Scaling** 3.1. Project Management and Learning; 3.2. Scale-up Phase 2.

4. **Conditional Emergency Response Mechanism**

1.2. Institutional arrangements for the implementation of the programme

i) Steering Committee

In accordance with Sections I.A.1 and I.A.5(a) of Annex 2 to the Financing Agreement, it is foreseen that a National Project Steering Committee (NPC) and a Provincial Project Steering Committee (PPPC) will be established for each participating province. These committees are to be created respectively by the order of the Minister in charge of Water Resources and Electricity and by orders of the Governors of the provinces and they will be responsible for: (1) defining the strategic orientations of the project; (2) approving the AWBPs and; (3) facilitate collaboration between the ministries and agencies involved in the project.
The meetings of the various steering committees will be held every six months or every time if necessary and the minutes of the meetings will be made public.

ii) Implementing agencies of the PASEA programme

The implementing agencies responsible for the implementation of the various activities of the PASEA are listed below:

- **At the national level** : the Water Project Implementation Unit, "CEP-O" in acronym, for project activities of national scope or concerning several provinces, the transfer of competences to the permanent provincial structures with the technical support of a national coordination team composed of the Sanitation Directorate "DAS", the National Office of Rural Hydraulics "ONHR", the Directorate of Hygiene and Public Health "DHSP") and the Directorate of Sanitation and Public Health (DHSP) and the Directorate of Sanitation Care and Partnership Establishments "DESP" of the Ministry of Public Health, Hygiene and Prevention (MSPHP), the Directorate of Education and Everyday Life "DEPC", the National Directorate of Construction, Maintenance, Rehabilitation and Management of School Infrastructure "DINAC" of the Ministry of Primary, Secondary and Technical Education (MEPST);
- **At the provincial level** : the Provincial Project Implementation Unit "UPEP" for activities at the provincial level, under the supervision of the CEP-O and the technical support of the Provincial Coordination Team composed of the provincial water service board and the provincial representations of the DAS, ONHR, DDSH, DSPD, etc. of the DINAC and the DEVC.

iii) Stakeholders and Targeted Actors

The Project's target actors are made up of:

- The populations living in the various Decentralized Territorial Entities, "ETD" in acronym, selected and those of the peri-urban areas of the cities selected in the provinces of Kasai, Kasai Central, Kasai Oriental and Kwilu;
- Schools and health centers in selected ETDs and cities;
- The population of different provinces selected from this first phase generally indirectly.

The project will be carried out with the involvement of the following stakeholders:

- the Prime Minister;

- The Ministries of Water Resources and Electricity, Environment and Sustainable Development, Primary, Secondary and Technical Education,
  Public Health, Hygiene and Prevention, Planning, Finance, Urban Planning and Housing and Land Affairs;
- The provincial governments of the selected provinces;
- Decentralized territorial entities (rural communes, sectors or chiefdoms) and cities in the chosen provinces;
- Public water and/or sanitation service operators (REGIDESO; ASUREP, private companies, etc.);
- ONHR, the provincial public water service boards;
- State bodies (ARSPE, OCE, etc.);
- Civil society organizations active in the project area; • Managers of social and educational institutions working in the project area
  (health centres, schools, markets, social promotion centres);
- Universities and vocational training centres.

For the two key sectors of the Project (water and sanitation), it is planned respectively:

(i) The provision of Technical Assistance for the establishment and strengthening of the Provincial Water Service Boards, including the development and adoption of a provincial governance framework for the management of service delegation contracts, the establishment of a fund for the maintenance and development of water supply, the installation and implementation of an Information Management System (IMS) for water operators and the preparation of related operational manuals for the water component and

(ii) The provision of Technical Assistance to the MEDD-DAS to develop a national SGI for rural sanitation and to support its implementation in the participating provinces for planning, regular SGI-based learning and revisions for the sanitation component.

Since September 2023, discussions have taken place within the Government for the gradual establishment of a real National Water-Hygiene-Sanitation (WASH) Platform with the aim of actively coordinating the WASH sector, monitoring and directing investments and optimizing transaction costs. It is expected that the WASH Platform will be operational by the end of 2024 or at the latest at the beginning of 2025. The Platform will have an important role to play in the collection of information related to the water and sanitation sectors. It therefore seems essential that the Task Force in charge of its implementation is well involved in the development of the Information Management Systems (IMS) that will be implemented through the Project.

Also, the Public Water Service Regulatory Authority (ARSPE) set up in 2022 will be able to be operational by the end of 2024. This agency will have an important role to play in the collection and processing of data, particularly on tariffs.

As a first step, it is necessary to diagnose the existing SGIs for both the water and sanitation sectors and to prepare the Terms of Reference to enable the Project to recruit one (or two) firm(s) that will develop one or two SGIs adapted for the drinking water sector and for the liquid sanitation and faecal sludge sector.

These Terms of Reference relate to the recruitment of an individual Consultant to diagnose existing SGIs in the water and sanitation sectors and to support the Project team in the

development of the Terms of Reference for the recruitment of the firm(s) responsible for developing and designing the new LMS(s) and assisting the PASEA project teams, the decentralised and decentralised authorities in the provinces concerned, the National WASH Platform and ARSPE in the implementation (implementation) of the new system(s).

2. OVERALL OBJECTIVE OF THE MISSION

The general objective is to have the Terms of Reference integrating the recommendations of a diagnosis of existing Information Management Systems (IMS) and other IMSs used in other countries in order to enable the recruitment of one or more firm(s) that will be responsible for developing and designing one or two new IMSs for the water and sanitation sectors.
The decision to proceed with the development of a single ToR (a single SGI for water and sanitation) or several ToRs (an SGI for water and an SGI for sanitation) will result from the diagnosis carried out.

3. SPECIFIC OBJECTIVES OF THE MISSION

The specific objectives of the mission are to:

- To have an inventory of the institutional mechanism and the coordination of actors and interventions in the water and sanitation sectors, including the future National WASH Platform, the Provincial Water Service Boards, municipalities, decentralized and decentralized specialized institutions, and possibly the Regulatory Authority of the Public Water Service (ARSPE);
- Carry out an exhaustive analysis of the availability and needs of data and indicators as well as their management by all the structures involved in the sectors;
- To provide a brief overview of regional and international experiences in SGI for the WASH sector (e.g. mWater, SIASAR, etc.). The (individual) Consultant will be responsible for ensuring that the experience of the sanitation regulator in Zambia is properly captured (including lessons learned at an early stage);
- Identify gaps in the information management of the water and sanitation sectors both at the national level and in the provinces concerned by the Project, taking care to consider future developments (e.g. City-Wide Inclusive Sanitation Plans - CWIS for sanitation);
- Produce a comprehensive comparison of the advantages and disadvantages (including financial) for one or more separate SGIs for the water sector and the sanitation sector. This judgement should be supported by feedback from other regions. Precise cost simulations (servers, updates) will be part of this comparison.
- Based on the decision taken by the CEP-O for one or two separate IMS, the (individual) Consultant will inventory the technical specifications and prerequisites for the development of new systems, or for the customization of existing systems (if applicable).
- In the event that two separate IMS are proposed, the (individual) Consultant will take care to specify through which channel the systems will be able to communicate
- Develop the Terms of Reference for the recruitment of the firm(s) in charge of designing and providing technical assistance for the implementation of one or two SGIs for the water and sanitation sectors.

To achieve the above stated objectives, the (individual) Consultant will be responsible for performing the following tasks:

✞ **Task 1** : Perform the diagnostic study of existing IMS and data needs

For this task, the consultant will have to carry out the following operations:

- Carry out an exhaustive diagnosis of the availability and needs of data and indicators in the water and sanitation sectors as well as all the structures involved in the various areas of water, hygiene and sanitation;
- To take stock of the institutional arrangements of the SGI (in particular the mWater solution implemented with the GiZ);
- Identify which actors feed the databases;
- Identify who and/or which institutions have access to the data;
- Analyze the strengths, weaknesses, opportunities and threats of the institutional set-up in its current structure, the coordination of actors and interventions in the different WASH sub-sectors in the provinces concerned by the Project;
- Identify the human, hardware, software and financial resource needs essential for IMS;
- Identify actions to improve data collection, processing, analysis and dissemination of information;
- Identify the technical capacity building needs of the actors who will be involved in the implementation of the IMS(s);
- Develop a roadmap for the implementation of the integrated database with an exhaustive and detailed budget (the budget will also be decisive to inform the Client's choice);
- To make relevant recommendations to guide the development of the computerized database.

✞ **Task 2** : Identify international and regional experiences and good practices

- Produce a brief overview of WASH SGIs used in other countries and highlight good practices (e.g. regarding the urban sanitation SGI in Zambia);
- If any SGIs have been abandoned in the DRC or other countries in the region, clearly indicate the reasons in the form of lessons learned.

✞ **Task 3** : Identify the standard data to be collected and imputed in the IMS

- Interviews with the Provincial Coordinators of the Provincial Water Service Boards in Kwilu, Kasai, Kasai Central and Kasai Oriental (and possibly in North Kivu);
- Interviews with experts working with ARSPE (MRHE) if they are already present at the time of the consultation;
- Interviews with the National Action Committee for Water, Hygiene and Sanitation (CNAEHA);
- Interviews with the Task Force in charge of the implementation of the National WASH Platform (if already effective at the time of the consultation);
- Interviews with experts from the National Directorate of Sanitation (MEDD) and with the Directorate of Hygiene and Public Health (DHSP / MSPHP) and their provincial representations;
- Interviews with municipalities in major cities in provinces or selected cities, after coordination with CEP-O;
- Production of a succinct report highlighting the key recommendations from these interviews.

✞ **Task 4** : Terms of Reference for Consultant/Firm in charge of the development of SGIs

- Transcribe the Client's choice to opt for several SGIs or harmonized SGIs into a single system (this decision will result from the diagnosis carried out);
- Transcribe the Client's choice in relation to the editing and access to the data of the SGIs;
- Specify the geographical location of the server that will host the data in accordance with the relevant law;
- Propose specifications and a realistic schedule;
- Specify how the TA should be mobilized to support the national agencies (ARPSE, DAS, DHSP, municipalities and the National WASH Platform) as well as the decentralized and decentralized provincial agencies (Provincial Public Water Service Boards, Office of Hygiene and Public Health, etc.).

4. EXPECTED RESULTS AND DELIVERABLES

The following table provides a summary of the expected results and planned deliverables:

Expected results	• Exhaustive diagnosis of the availability and needs of data and indicators of the structures benefiting from the Project; • SWOT analysis: strengths, weaknesses, opportunities and threats of the institutional set-up in its current structure; • Overview of international experiences with EHA SGI and summaries of good practices; • Recommendations from interviews with strategic actors (National Platform for Human Rights, Municipalities, RPSPE, MEDD, MRHE, etc.); • Proposal of various options and configuration: a single IMS for the two sectors or separate IMS with a communication and exchange channel between the different systems; • The human, material and financial resource requirements (including a budget for the various options proposed) essential for the management of the computerized database are assessed; • Improvement actions in terms of data collection, processing, analysis and dissemination of information are defined; • The technical capacity building needs of the actors who will be involved in the implementation of the database are identified; • A roadmap for the implementation of the integrated database with a detailed and comprehensive budget (the budget will be decisive for the selection of options) is developed; • The RDTs are developed. • A formulation of relevant recommendations to guide the development of the computerized database.

Deliverables	• Inception report including the methodology and work plan that will be used by the (individual) Consultant to carry out his/her mission; • Diagnostic report including international experiences and good practices. This report should provide relevant recommendations (to inform the selection of a sector-specific IMS or a single EHA IMS, a decision to be made by the Client) and contain a roadmap with estimated budgets for the implementation of the IMS ; • The ToR(s) of the firm(s) in charge of the Design and Technical Assistance (TA) for the implementation of the SGI(s) for the water and sanitation components in the provinces concerned by the PASEA Project.

Reports should include updates on milestones, issues encountered, and any changes to the timeline.
A workshop will be organized to allow relevant stakeholders to validate the recommended SGI option. All costs of organizing the workshop will be borne by the project.

5. DURATION OF THE MISSION

The duration of the mission is ninety (90) calendar days. The main steps are as follows with the number of days:

- Production of the methodological note and the work plan: **10 days**
- Diagnosis of existing IMS, overview of international experiences and production of the roadmap: **M+ 75 days**
- Production of the terms of reference for the recruitment of the firm(s) in charge of the design and technical assistance for the implementation of the SGI(s) for the water and sanitation components in the provinces concerned by the PASEA Project: **M+90 days**

7 EXECUTION OF THE MISSION

7.1 Organization

The (individual) Consultant will work closely with the CEP-O and the various institutions mentioned above. However, the Committee will take full responsibility for the analysis and interpretation of the data obtained, as well as for the conclusions and recommendations of its reports. It will take into account the various remarks and comments of the departments directly involved in the project on its provisional reports.

The (individual) Consultant will be expected to apply a participatory approach. It is a question of involving all the stakeholders concerned by these services. The (individual) Consultant must be based in the DRC.

7.2 Consultant Profile

- A master's degree and/or diploma (Bac + 5) in computer science, data management, IMS, data analytics, information technology or another discipline related to information management systems.

- Have at least 10 years of experience in data administration (Data collection, data cleaning and validation, data visualization, ...). The consultant must demonstrate proficiency in KoboCollect, ODK and similar software to create data collection tools, mastery of transforming and rationalizing data from one digital format to another (e.g., from CSV formats to formats supported by SQL; between KoboCollect and structured data tables, etc.);
- Have carried out at least 2 similar missions (Diagnosis or Design and implementation of SGI) in the last 5 years. Experience in the water and/or sanitation sectors is preferred);
- Experience working with ArcGIS, QGIS and Microsoft Excel, SPSS, Stata, Google Data Labs, PowerBi, ...;
- Fluency in French is compulsory and English is an asset;
- Any other relevant qualifications.

The (individual) Consultant must also have the following qualifications:

- Ability to communicate, negotiate, develop and maintain strategic partnerships in the field of research and data management;
- Ability to produce high-quality work while meeting multiple deadlines simultaneously;
- Innovative, self-motivated, and able to work with minimal supervision.

8 RESPONSIBILITY OF THE CEP-O

The Consultant (Individual) will be recruited by the CEP-O in accordance with the current World Bank guidelines for the mobilization of consultant services based on the validated Terms of Reference.
The (individual) consultant will work under the supervision and regular monitoring of the CEP-O and the approval of files will be the responsibility of the CEP-O.
The CEP-O will inform the Consultant of the CEP-O persons who will be assigned to monitor the progress of the mission and will make every effort to facilitate the Consultant's access to existing documents and any other information necessary for the success of the mission.

9 LOGISTICS

The (individual) Consultant will cover the costs of his field trips and all the necessary means (mobile phone, internet connection, computers, printers and various consumables and all other equipment deemed useful including accommodation) allowing him to perform his services in conditions of efficiency.

10 KICK-OFF MEETING

At the start of the mission, a meeting will be held between the (individual) Consultant, the CEP-O, in order to agree in particular on:

- Any amendments to be made to the work plans to meet the terms of reference;
- The technical approach and methodology of the individual Consultant and his work program for the realization of the mission;
- Organizing collaboration with CEP-O and other stakeholders;
- The organization of any visits to the different project sites;
- communication during the mission with the CEP-O in the provinces concerned by the project.

11 GENERAL CONDITIONS:

The (individual) Consultant will comply with all standard legal and procedural terms, including confidentiality agreements, ownership of work products and conflict of interest policies.

During the term of the Services, the (Individual) Consultant will not disclose any proprietary or confidential information regarding the Services without first obtaining the Client's permission.

Stakeholder Interview Guide for the Development of the Terms of Reference for the Acquisition of a Management Information System(s) (MIS) on Water, Hygiene and Sanitation (WASH) Aspects for the four targeted provinces of the PASEA Project: Kwilu, Kasai, Kasai Central and Kasai Oriental

This guide presents the key themes around which the stakeholder interview to be consulted for the development of ToRs for the acquisition of one or more SGI systems related to water and sanitation in the target provinces of the PASEA project is structured

It facilitates the deployment of the more detailed questionnaire on ODK for the consolidation of data in electronic format for diagnostic analyses.

MAINTENANCE AND REPORT COLLECTION

1. **Operating Context and Objectives**
   - What are the main objectives and missions of your organisation (structure) in the water or sanitation sector?
   - What are your current and future strategic priorities?
   - Who are the main stakeholders you interact with (governments, existing partners, NGOs, private sector)?
2. **Actors and Stakeholders**
   - Who are the key stakeholders in your organization? What types of data do you share with them?
   - How do you coordinate your actions with other players in the sector?
   - Are there shared governance or consultation processes for data management?

- How satisfied are you with the coordination and sharing of information between the different stakeholders? What are the points of improvement for you?

3. **Data Collection and Sources** • What types of data do you currently collect (water quality, infrastructure, service coverage, etc.)?
   - What are the main sources of this data (field surveys, sensors, agency reports)?
   - How often are they updated?
   - Who is responsible for collecting and validating this data?
   - How do you plan to improve your approaches to data collection and source management? (Gaps in the data collected now, etc.)
4. **Data Management and Quality**
   - What data storage and management methods do you use?
   - What standards or protocols are applied to ensure data quality?
   - Do you face specific challenges related to data quality or accuracy?
5. **Current Tools and Infrastructure** • What IT systems or tools do you use for information management?
   (Example: databases, GIS, mobile applications)
   - How reliable and accessible are these systems?
   - What problems or limitations do you encounter with these tools?
6. **Data Analysis and Use**
   - What analyses are you currently conducting for decision-making (e.g., monitoring of indicators, forecasts)?
   - What types of reports do you generate and how often?
   - Is the data used for monitoring, evaluation, planning, or other functions?
   - How satisfied are you with the current analyses and data? What are the points of improvement for you?
7. **Data Exchange and Sharing**
   - Which organizations do you share data with and how?
   - Are there any data-sharing platforms or interoperability protocols?
   - Do you encounter any legal, technical or organizational barriers to data sharing?
8. **Needs and Expectations**
   - What do you think are the main needs to be met to improve information management in your sector?
   - What features or capabilities would you like to see in a new information management system?
   - What are the key success indicators for you?
9. **Skills and Resources** • Do you have a dedicated data management and analytics team?
   - What skills are missing or need to be strengthened for better information management?
   - Do you have a budget or resources available for adopting new systems?
10. **Data Security and Privacy**
   - What security measures do you apply to protect sensitive data?
   - What are some of the data privacy challenges you've faced?
   - What compliance policies do you follow when it comes to data?